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PROMISE: A pre-trained knowledge-infused multimodal representation learning framework for medication recommendation

Jialun Wu [a], Xinyao Yu [b], Kai He [b,*], Zeyu Gao [c], Tieliang Gong [a]

[a] School of Computer Science and Technology, Xi'an Jiaotong University, Xi'an, Shaanxi, China
[b] Saw Swee Hock School of Public Health, National University of Singapore, 12 Science Drive 2, Singapore
[c] Department of Oncology, University of Cambridge, United Kingdom

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ABSTRACT

Electronic Health Records (EHRs) significantly enhance clinical decision-making, particularly in safe and effective medication recommendation based on complex patient data. Current methods, while encoding each medical event individually with domain-specific knowledge, inadequately harness multi-source domain knowledge and neglect the interrelations among various medical codes, the influence of historical patient visits, and the relevance of similar patient trajectories. To address these limitations, we present PROMISE, a multimodal medication recommendation framework that progressively learns patient representations from specific health states to a comprehensive view. PROMISE integrates domain knowledge into modality-specific encoders to improve local and global patient representations, facilitating enhanced medication recommendations through the interaction of patient representations from various modalities. Specifically, within the code modality, PROMISE utilizes encoding of EHR hypergraphs to learn patient representations featuring structured information. Simultaneously, in the text modality, it acquires patient representations with semantic information by encoding clinical texts obtained from tables. Our framework surpasses state-of-the-art baselines with up to 2.06% and 1.97% improvements on key metrics within the MIMIC-III and IV datasets, respectively, confirming its effectiveness and superiority.

1. Introduction

The escalating prevalence and swift accumulation of Electronic Health Records (EHRs) offer substantial opportunities for researchers to construct data-driven models, thereby potentially improving the quality of personalized healthcare (Choi, Bahadori, Schuetz, Stewart and Sun, 2016; Li et al., 2017; Ma et al., 2017, 2020). Medication recommendation, a critical aspect of healthcare, aims to predict safe and effective medications personalized to patients with complex health conditions (Wu, Dong, Gao, Gong and Li, 2023; Wu, He, Mao, Li and Cambria, 2023; Wu, Qian et al., 2022). As shown in Fig. 1, EHRs contain comprehensive information, including diagnoses, procedures, and medication records. Effectively analyzing this data from EHRs can significantly contribute to informed clinical decision-making.

Apart from *Rule-based methods* that extract knowledge and patterns from raw data, current methods can be categorized into two groups: *(1) Instance-based methods*, which focus on predicting medication combinations solely based on the patient's current hospital visit (Gong, Wang, Wang, Wang, & Liu, 2021; Zhang, Chen, Tang, Stewart, & Sun, 2017). Nevertheless, these methods tend to disregard temporal dependencies within patient visits. To address this shortcoming, *(2) Longitudinal-based methods*, which utilized

* Corresponding author.
E-mail address: KAI_HE@nus.edu.sg (K. He).

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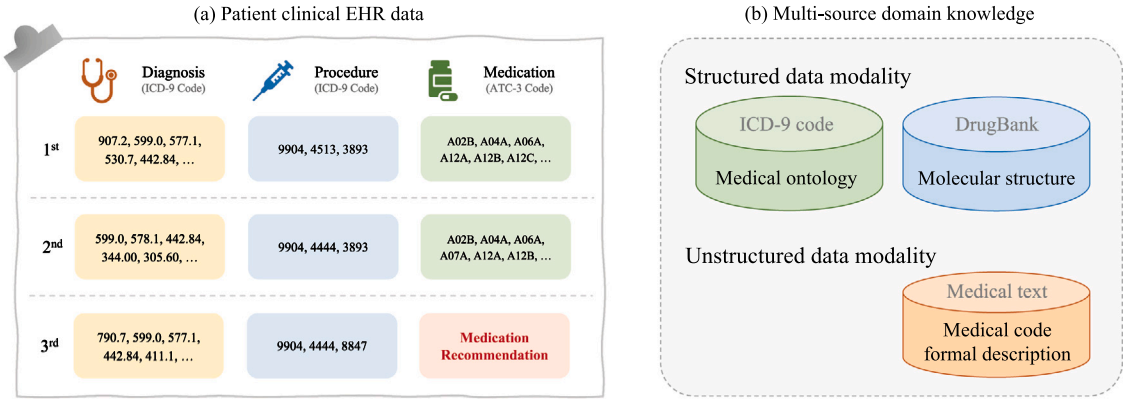


Fig. 1. Concrete description of medication recommendation task. (a) Patient clinical EHR data, (b) Multi-source domain knowledge.

historical visit records to augment patient representation and facilitate medication recommendations. These methods primarily focus on exploring the hidden patterns within EHRs (Choi, Bahadori, Sun et al., 2016; Shang, Xiao, Ma, Li and Sun, 2019; Wang, Chen, Pi, & Yue, 2021; Wu, Qiu, Jiang, Qi and Wu, 2022; Yang, Xiao, Glass, & Sun, 2021). Additionally, some methods integrate external knowledge to learn medication representations, which further enhances model performance (Yang, Xiao, Ma, Glass and Sun, 2021; Yang, Zeng, Wu, & Yan, 2023). Despite their importance, previous studies exhibit the following limitations:

(1) Domain knowledge underutilization. While many methods integrate biomedical knowledge to improve prediction accuracy (Shang, Ma, Xiao and Sun, 2019; Yang, Xiao, Ma et al., 2021; Yang et al., 2023), they often focus on single-source domain knowledge related to specific medical events (Choi, Bahadori, Song, Stewart, & Sun, 2017; Ma et al., 2019, 2018). These approaches overlook the potential predictive value inherent in richer, multi-source domain knowledge, which covers a wider range of medical events and could greatly improve the precision of clinical predictions and treatments. Moreover, current studies often concentrate solely on processing EHRs in tabular form, a single modality, thus overlooking the rich insights that can be gleaned from multimodal data sources. Consequently, we argue that a framework that encompasses a comprehensive multi-source domain knowledge, covering all medical events in EHRs.

(2) Inadequate representation learning. Previous studies often encode medical events from a patient's current visit independently and then combine them for prediction (Shang, Xiao et al., 2019; Wu, Qiu et al., 2022). However, such approaches overlook the connections between different medical events and the influence of past visit information on present health states. Moreover, these methods usually concentrate solely on the target patient, disregarding the influence of similar patients' medical histories, which is critical in clinical decision-making (Suo et al., 2017; Wu et al., 2021; Zhang, Wang, Hu, & Sorrentino, 2014). Additionally, there are few studies that utilize multimodal clinical information for patient representation learning, which indicates a significant opportunity to leverage a wide array of patient data, encompassing both structured and unstructured information, for developing nuanced and precise patient representations. Thus, it is essential to utilize the relationships within multifaceted information for effective and comprehensive patient representation learning.

To overcome the aforementioned limitations, we propose PROMISE, a **PR**e-trained kn**O**wledge-infused **M**ult**I**modal repre**SE**ntation learning framework for medication recommendation. PROMISE integrates advanced information processing technologies from the biomedical field. These technologies encompass initializing code embeddings with multi-source domain knowledge, fusing various modality-driven learning flows to augment patient representation, and employing multiple loss functions during pre-training to achieve robust representation.

Specifically, to *fully utilizing multi-source domain knowledge*, PROMISE introduces a multimodal framework with various modality-driven learning flows, each integrating domain knowledge and employing modality-specific methods. While as illustrated in Fig. 1.(b), the term "multi-source domain knowledge" refers to the diverse types of information from various domains, such as medical records and biological databases, and "multimodal" pertains the ability of our framework to process and integrate this information across different data formats and structures. In the code-level representation learning flow, given that domain knowledge typically manifests in graph structures (*e.g.*, ontologies and molecular graphs), PROMISE encodes the nodes within these graphs and the information flows, resulting in code initialization embeddings enriched with structural details. In the text-level learning flow, PROMISE converts tabular EHR data into multiple sentences, subsequently replacing medical codes with their formal descriptions, creating an unstructured medical record. This transformation facilitates the learning of code embeddings with implicit semantic information. These strategies enable PROMISE to effectively utilize multi-source domain knowledge from varied modality-driven learning flows, and further improve the ability of patient representation and the accuracy of medication recommendations.

To *attain comprehensive patient representations*, PROMISE employs diverse representation learning methods within and across different learning flows. Drawing inspiration from the physicians' diagnostic process, PROMISE progressively learns patient representations from specific instances to a holistic view. "Local health state" refers to the patient's condition within a specific timeframe,

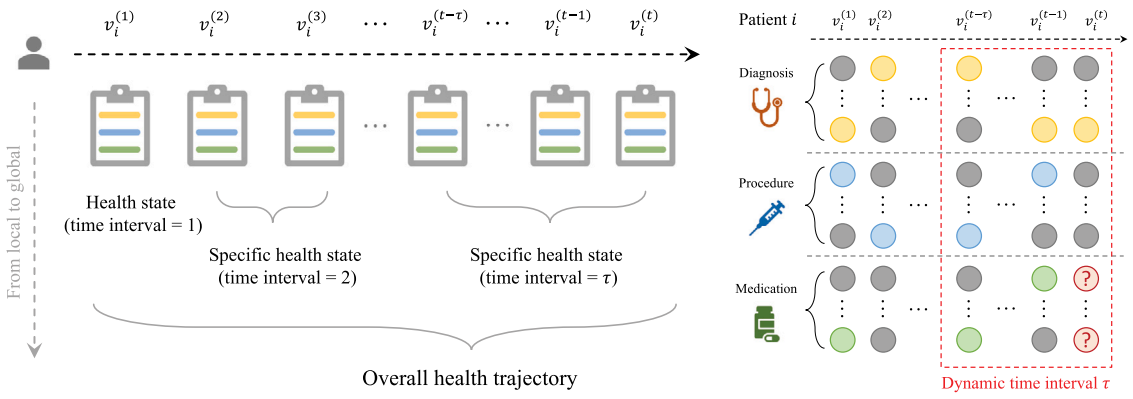


Fig. 2. Concrete and symbolic descriptions of patient health state from local to global. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

identified through the segmentation of complex health states in time-series data into subsequent intervals. These local states are initially learned and then aggregated to depict the patient's overall health progression, embodying the transition from specific (local) health conditions to a generalized (global) health state of the patient over time. In the code-level learning flow, PROMISE utilizes sub-hypergraphs to encode multiple hyperedges within predetermined time intervals, capturing local state representations that include historical and structural correlation information. Meanwhile, in the text-level learning flow, PROMISE utilizes pre-trained language model to encode multiple sentences within predetermined intervals to capture historical and semantic information in patient health state representations. Additionally, PROMISE introduces a multi-level attention mechanism to integrate trajectory data from similar patients to obtain global patient representations of different modalities. This global health state encapsulates the comprehensive health trajectory of a patient, integrating insights from various local states. Finally, PROMISE provides detailed insights into a patient's health status by correlating multimodal representations and maintaining single-modal abilities, further enhanced through pre-training with both modality-specific and cross-modality losses to develop comprehensive patient representations. Fig. 2 shows the patient representation learning process of PROMISE from local to global.

The contributions of our framework are summarized as follows:

- Our framework integrates multi-source domain knowledge. It initialized medical code embeddings with structural information by analyzing graph-structured external knowledge. Concurrently, it encodes unstructured records derived from patient tabular data to capture patient representations with semantic information.
- We present a new multimodal representation learning framework. This framework initially learns local state representations for each time interval in a patient's time series data and subsequently acquires a global representation via similar patient trajectories. The framework attains comprehensive patient representation by enabling interactions across different modalities while maintaining modality-specific capabilities.
- Our study involves extensive experiments using the publicly accessible MIMIC-III and MIMIC-IV datasets. PROMISE outperforms existing state-of-the-art benchmarks, showing improvements of 1.61%, 1.64%, and 2.06% in Jaccard, F1, and PRAUC score on MIMIC-III dataset, and 1.97%, 1.42%, and 1.14% on MIMIC-IV dataset, respectively. These advancements are accompanied by decreased ADDI rates and increased SDDI rates, indicating enhanced safety and efficacy in medication recommendation.

2. Related works

2.1. Medication recommendation

The prevalent adoption of EHRs in healthcare facilities has encouraged researchers to employ deep learning techniques to derive meaningful information from EHR data, aiming for individualized and accurate healthcare solutions. This involves using the patient EHR data (as shown in Fig. 2), denoted as i, is depicted as the medical record sequence, $\mathrm{V}_i = [v_i^{(1)}, v_i^{(2)}, \ldots, v_i^{(t)}]$. Each visit in this sequence includes various medical events (*i.e.*, the diagnosis, procedure, and medication). This component is designated as 1 (represented by a colored point) if the medical code is included in the tth visit; conversely, it is assigned a value of 0 (indicated by a gray point) in its absence. A red box highlights the visits within a dynamic time interval τ, specifically from $v_{t-\tau}$ to v_t. However, the complexity of EHRs, containing detailed patient medical histories and health information, poses challenges in achieving accurate data analysis and interpretation.

The development of early warning systems represents a major application of EHR data mining. Leveraging obscured patterns in patient medical histories to forecast upcoming health events. A primary challenge in this context is the accurate alignment of complex patient health statuses with relevant medical codes (Li et al., 2021). Achieving this necessitates the creation of informative patient representations, the efficient encoding of medical codes, and the integration of various domain knowledge sources.

In the recent review paper (Ali et al., 2023; Etemadi et al., 2023), prescribing medications in response to a patient's diagnosis presents a complex challenge. Early works in medication recommendation have explored **Rule-based methods** that extract knowledge from EHR data. UIMA (Solt & Tikk, 2009) can identify patterns and extract rules that relate patient characteristics to appropriate treatments. UCT (Lakkaraju & Rudin, 2017) learns the hidden pattern between patient representation and treatment plans with utilizing the Markov decision processes (MDPs). Zhao et al. (2020) combined clinical guidelines with medical data to construct a decision tree for and treatment recommendation. However, these approaches can lead to conflicting rules and face challenges in generalization and scalability.

Most current healthcare application techniques primarily depend on RNN (Hochreiter & Schmidhuber, 1997) and attention mechanisms (Vaswani et al., 2017), owing to their enhanced ability to process time-series EHR data. Generally, these methods fall into the following two distinct categories.

When concentrating on evaluating a patient's current health status, the *Instance-based methods* focus on assessing the patient health condition and approach medication recommendation as a process of sequential decision-making task. For example, LEAP (Zhang et al., 2017) provides the medication recommendation with utilizing a content attention mechanism in the sequence-to-sequence manner during a patient's current visit. Gong et al. (2021) embed patient demographics and diagnoses into a compact space for medication recommendation, treating it as a link prediction issue that accounts for diagnoses and adverse reactions. Wang, Zhang, He, and Zha (2018) combine demographics, diagnoses, and medication data using a linear method. However, all these approaches ignore overlook past visit diagnoses, impacting accuracy and personalization.

Different from the instance-based method, the **Longitudinal-based methods** capitalize on the temporal correlations present in a patient's historical health records (Wang, Cai, Yang and Mei, 2022). RETAIN, as a prominent example, which employs a dual-level attention mechanism to identify the significant previous visits and critical clinical variables (Choi, Bahadori, Sun et al., 2016). However, these approaches frequently overlook crucial safety considerations, such as predicting and preventing the prescription of medications that could lead to adverse side effects for the patient.

To enhance safety in medication recommendation, GAMENet integrates drug–drug interaction (DDI) by simultaneously examining patient records and the construct DDI graph, further offering a comprehensive analysis (Shang, Xiao et al., 2019). SafeDrug advances this by extracting and encoding molecular structure information, thereby improving the safety of medication recommendations (Yang, Xiao, Ma et al., 2021). Furthermore, 4SDrug aims to ensure safety by recommending a limited number of drugs (Tan et al., 2022). In a more targeted approach, MoleRec employs a hierarchical architecture to associate medication substructures at the molecular level with specific diseases, facilitating personalized medication recommendations (Yang et al., 2023). Some recent studies concentrate on predicting the probability of medication repetition with considering the patient's past visit history. For example, MICRON utilizes recurrent residual networks to track medication changes across different visits (Yang et al., 2021). COGNet, introducing a copy module, have redefined the approach to medication recommendation by framing it as a medication change prediction issue (Wu, Qiu et al., 2022).

2.2. Multimodal recommendation systems

As the internet and multimedia data flourish, multimodal recommendation systems have become a key area of research (Wang, Zhang, Wang and Ricci, 2022; Zhu et al., 2021), integrating various structured and unstructured data types such as text, images, and tables to provide more accurate and customized suggestions (Jujjavarapu et al., 2023). These systems primarily use advanced deep learning techniques to efficiently process complex multimodal data, thereby more comprehensively capturing user preferences (Liu, Mao, Geng, & Cambria, 2023; Mu & Wu, 2023). Moreover, they address challenges like data scarcity and the cold-start problem in recommendation systems by utilizing the abundant multimodal information (Cai, Qian, Fang, Hu, & Xu, 2023; Tang, Yang, & Wang, 2023).

These approaches generally include extracting features from different modalities and facilitating the interaction and recommendation of features across these modalities. Multi-modal data is initially processed using specific modality encoders, like vision transformers for images and BERT for text (Han et al., 2022; Kenton & Toutanova, 2019). While standard recommendation models determine probabilities using this fused data, they often struggle with data scarcity. Therefore, numerous studies suggest improving representation through the integration of multi-modal information.

Multimodal recommendation systems are widely applied in time series analysis across various sectors (Yi et al., 2023). In finance, they analyze market data and news to provide investment advice (Cheng, Yang, Xiang, & Liu, 2022); for researchers, they offer a curated list of references for specific manuscripts (Cai, Wang, Yang, & Mei, 2022; Mei et al., 2022); and in healthcare, they combine medical records with diagnostic imaging (Islam, Nooruddin, Karray, & Muhammad, 2023; Muhammad et al., 2021). Overall, these systems enhance precision and personalization in services across different industries by integrating multiple data types and advanced processing techniques.

2.3. PROMISE in relation to exiting methods: similarities and differences

PROMISE distinguishes itself from traditional medication recommendation methods by employing a novel multimodal framework that learn patient representation from separate health states to a comprehensive view. The first notable distinction of PROMISE lies in its unique ability to incorporate multi-source domain knowledge, addressing a significant gap in existing methodologies. Traditional approaches often depend on a constrained set of data to individually code medical events, thereby failing to leverage the breadth of available knowledge comprehensively. In contrast, our framework integrates a wider array of domain knowledge, including formal

textual descriptions of medical codes, diagnostic code ontologies, molecular graphs, and drug–drug interaction (DDI) relationships, offering a more detailed understanding of a patient's health status.

Moreover, PROMISE's innovative approach to modeling the relationships between different medical events within a patient's visit across time frames enables the learning of local health states more effectively. It then synthesizes these into a global representation of the patient by considering historical visit data and insights from similar patient trajectories. This capability directly addresses the overlooked gap in existing research, which often simplifies the complex interplay and distinctions between independent clinical events, overlooking the intricate dynamics of medical interactions. Such conventional methods inadequately capture the complexity of patient health states and ignore the valuable context provided by historical and comparative patient data, critical for informed clinical decision-making.

By bridging these gaps, PROMISE not only enhances the learning of local representations but also constructs a comprehensive global view of patient health, substantially improving the accuracy and efficacy of medication recommendations. These distinguishing features position PROMISE as a significantly advanced framework in the field of medication recommendation, offering a clear departure from and advancement over existing methodologies.

3. Research objectives and research questions

This study aims to evaluate the effectiveness of PROMISE, an innovative multimodal representation learning framework, aiming at improving medication recommendation processes. The core objective is to demonstrate how the integration of multi-source domain knowledge and the encoding of complex EHRs data can significantly improve the accuracy and comprehensiveness of medication recommendations. Specifically, the research seeks to understand the impact of leveraging diverse sources of domain knowledge to enrich medical codes with structural and semantic information, thereby facilitating a more nuanced understanding of patient health states from both local and global perspectives.

Additionally, the study will investigate the effectiveness of PROMISE's modality-specific learning methods in accurately modeling the complex interrelations among various medical events and their impact on patient health outcomes. By analyzing patient trajectories, the research aims to construct a global representation of a patient's health status, offering deep insights into their overall health condition and the dynamics of their medical history.

A key question driving this research is how the sophisticated methodologies employed by PROMISE, including its ability to integrate and analyze data from multiple modalities, contribute to the framework's success in providing precise and effective medication recommendations. Moreover, the study will explore the improvements in recommendation accuracy brought about by enhanced cross-modal interactions and the maintenance of modality-specific capabilities, underpinned by modality enhancement modules and pre-training tasks.

By addressing these aspects, the research not only aims to validate the benefits of using the PROMISE framework for medication recommendations but also to contribute to the broader field of medical informatics by offering insights into the effective integration of multimodal data for patient care optimization.

4. Method

4.1. Preliminaries and task formulation

Preliminaries: As depicted in Fig. 1.(b), patient longitudinal EHR data comprises a series of diverse medical events, such as diagnoses, procedures, and medications. These events are encoded using a specific medical coding system, represented by sequences of medical codes. Diagnosis codes are denoted as $d_1, d_2, \ldots, d_{|D|}$, procedure codes as $p_1, p_2, \ldots, p_{|P|}$, and medication codes as $m_1, m_2, \ldots, m_{|M|}$. Here, D, P, and M represent the sets of diagnosis, procedure, and medication codes, respectively. $|D|, |P|, |M|$ indicate the total count of distinct medical codes in each category. The notation $|\cdot|$ refers to the cardinality of a set. For ease of reference, we collectively refer to these medical codes as $c_{|C|}$, where C encompasses all categories of medical codes.

Each patient in the dataset is represented as a sequence of visits, denoted by $\mathrm{V}_i = [v_i^{(1)}, v_i^{(2)}, \ldots, v_i^{(T_i)}]$, where i belongs to the set $\{1, 2, \ldots, \mathrm{N}\}$, with N being the total number of patients and T_i representing the number of visits for the ith patient. The tth visit of the ith patient is expressed as $v_i^{(t)} = [d_i^{(t)}, p_i^{(t)}, m_i^{(t)}]$. Here, $d_i{}^{(t)} \in \{0,1\}^{|D|}$, $p_i{}^{(t)} \in \{0,1\}^{|P|}$, and $m_i{}^{(t)} \in \{0,1\}^{|M|}$ are multi-hot vectors representing the presence or absence of diagnosis, procedure, and medication codes, respectively, for that particular visit. Meanwhile, the health state trajectory of patient i from his/her first visit $v_i{}^{(1)}$ to the τth visit $v_i{}^{(\tau)}$ is denoted as $V_i{}^{1:\tau}$.

Task formulation: The task of medication recommendation is framed as a multi-label prediction problem (Gong et al., 2021; Shang, Xiao et al., 2019), where historical patient visit records are utilized to predict the medication combinations for upcoming medical visits. Employing the notations previously defined, our framework aims to learn a function $f(\cdot)$ that proposes medications at each time-step t for various patients. The function $f(\cdot)$ can predict medications $\hat{m}^{(t)} \in \{0,1\}^{|M|}$ based on current and past visit records, as denoted by the equation:

$$\hat{m}^{(t)} = f([d^{(i)}]_{i=1}^{t}, [p^{(i)}]_{i=1}^{t}). \tag{1}$$

Our goal is to align the predicted medication $\hat{m}^{(t)}$ as closely as possible with the actual prescription $m^{(t)} \in \{0,1\}^{|M|}$.

The important notations and its description utilized in this paper are listed in Table 1. In the subsequent sections of this paper, we will exclude the subscript i when its context and meaning are evident and unmistakable.

Table 1
The utilized notations in PROMISE.

Module	Utilized notation	Description
Raw medical codes	$v_i^{(1)}, \ldots, v_i^{(T_i)}$	Patient i's health states V_i in each visit
	$d_i^{(t)}, p_i^{(t)}, m_i^{(t)}$	Patient i's clinical visit record at tth visit $v_i^{(t)}$
	D, P, M	Diagnosis, Procedure, and Medication set
	$V_i^{1:\tau}$	Patient i's health trajectory from $v_i^{(1)}$ to $v_i^{(\tau)}$
Modality-specific encoding	e_c^{uni}, e_c^{inh}	The unique and inherent embeddings of medical code c
	$nb(\cdot), ch(\cdot), pa(\cdot)$	The connected neighbor, child, and parent nodes set
	$F_{\text{Agg}}(\cdot,\cdot,\cdot)$	The embedding aggregation function (GAT)
	$\mathcal{G}^s, \mathcal{G}^a, \mathcal{G}^o$	The knowledge of SDDI, ADDI, and medical ontology
	X_d, X_p, X_m	The diagnosis, procedure, and medication embeddings
	$\mathcal{HG} = (\mathcal{V}, \mathcal{E})$	The constructed patient hypergraph
	q_C^L, q_T^L	The patient local state representations from two learning flows
Multimodal learning	q_C^G, q_T^G	The patient global state representations from two learning flows
	$\alpha_i^{(t)}, \beta_i^{(t)}$	The trajectory-level and patient-level attention
	$R_{\text{Code}}, R_{\text{Text}}$	The augment code-level and text-level patient representation
Pre-train and fine-tune	$\widehat{q}_i^{(t)}, \widetilde{q}_i^{(t)}$	The patient representations $q_C^{(t)}$, $q_T^{(t)}$ in shared feature space
	$o^{(t)}, \hat{o}^{(t)}$	The predict probability and medication combination at visit t

4.2. Architecture of PROMISE

In this section, we describe in detail the architecture of PROMISE, which is specifically designed for the task of medication recommendation.

4.2.1. Framework overview

PROMISE effectively integrates multi-source domain knowledge within a multimodal learning framework, aiming to learn patient representations from local to global perspectives, thereby comprehensively analyzing the clinical data and further prescribing medication combination. As shown in Fig. 3, the overview architecture of PROMISE is divided into three distinct stages:

- **Modality-specific local representation learning:** As shown in Fig. 3.(a), the first stage utilizes patient clinical data as input, integrates multi-source domain knowledge into dual learning flows (*i.e.*, code-level and text-level), and further encodes to obtain patients local state representations in different modalities, respectively. The *code-level learning flow* initially acquires medical code embeddings enriched with structural domain knowledge, followed by the construction and encoding of the EHR hypergraph to discern high-order patient state representations. Additionally, the *text-level learning flow* amalgamates unstructured medical text with EHR tabular data and domain knowledge, aiming to encode semantic patient state representations.
- **Multimodal global representation learning:** As shown in Fig. 3.(b), the second stage focuses on learning patient representations from local to global in each modality, leveraging the correlation between multiple modalities to enhance these representations. It incorporates a similar patient retrieval module for fine-grained trajectory-based similarity measurements, alongside a global representation learning module employing multiple level attention mechanisms. Additionally, the multimodal enhancement module ensures the interaction of global patient representations across different modalities while preserving the integrity of unimodal representations.
- **Pre-training and Fine-tuning:** As shown in Fig. 3.(c), the last stage aims at acquiring accurate medication results with comprehensive patient representations. This stage redefines the pre-training module for multimodal application, which integrates multiple information-constrained pre-training tasks, such as modality-specific prediction and cross-modality learning, facilitating effective knowledge fusion. The fine-tuning module employs feed-forward neural networks to actualize the medication recommendation.

Each component of PROMISE is and described sequentially below.

4.2.2. Modality-specific local representation learning

The EHRs utilized in clinical practice are characterized by their multimodal nature, including both structured-data (*e.g.*, time series visit records, tabular EHR data) and unstructured-data (*e.g.*, text data). At this stage, we establish code-level and text-level learning flows, subsequently encoding patient local state representations with respective structural and semantic information.

Specifically, Fig. 4.(b–c) concentrate on acquiring the structural representation of patients by leveraging structured medical codes and hierarchical domain knowledge. Meanwhile, Fig. 4.(d–e) illustrate our approach to constructing unstructured medical health records using formal descriptions derived from various codes, thereby enabling the extraction of patients' semantic representation information.

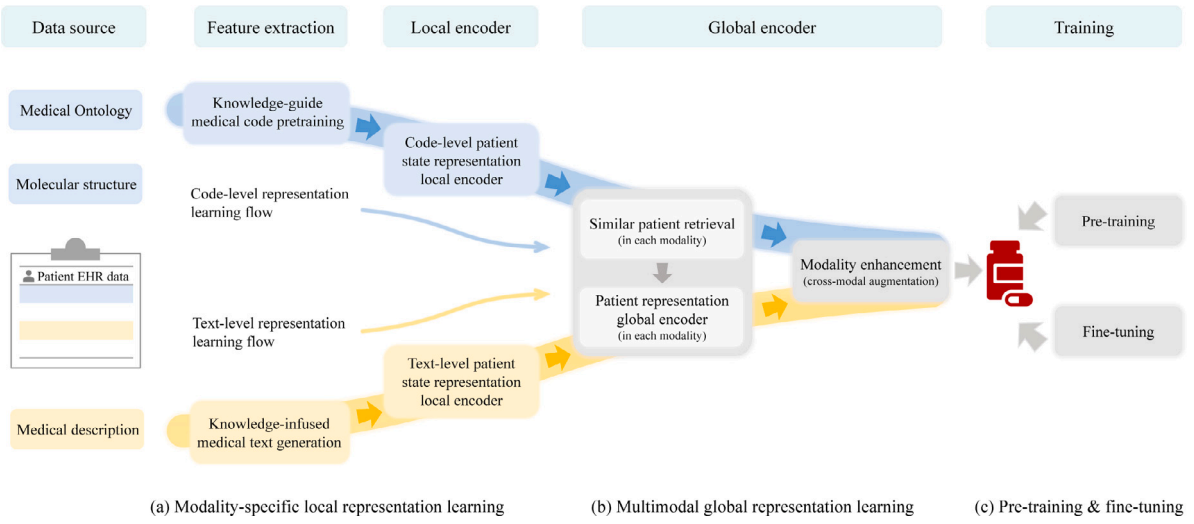


Fig. 3. The overview of the proposed framework, PROMISE. The input of PROMISE are the patient clinical EHRs data and multi-source domain knowledge. The output of PROMISE is the predicted medication combination. Our PROMISE consists of three stages, (a) Modality-specific local representation learning stage, (b) Multimodal global representation learning stage, and (c) Pre-training and fine-tuning stage.

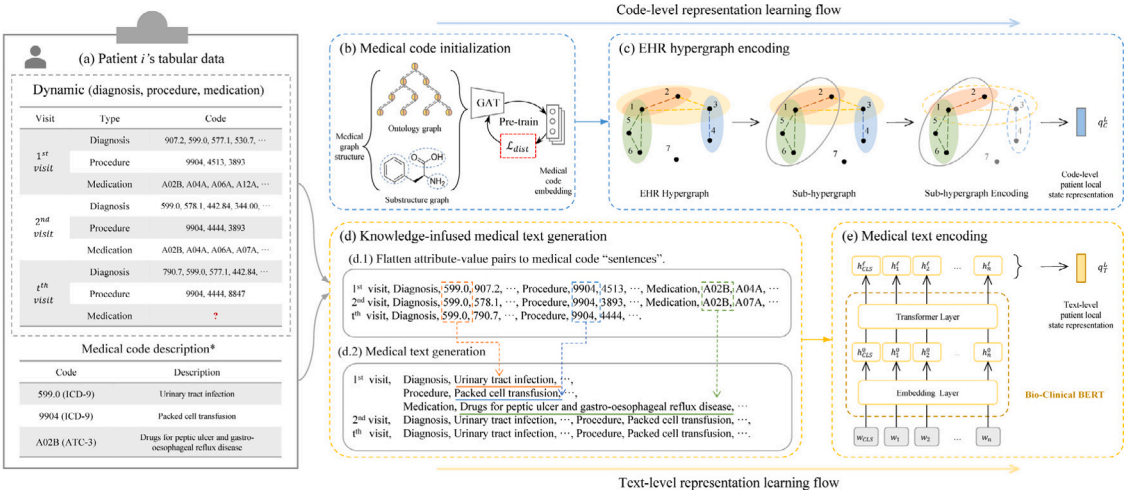


Fig. 4. The architecture of the Modality-specific local representation learning stage. The input to this stage is the multi-source and multimodal domain knowledge and patient tabular EHR data. The output of this stage is the code-level and text-level patient local state representations in different level representation learning flow, respectively. In this figure, PROMISE consists of the following components, (a) Patient *i*'s EHR tabular data (sample), (b) Medical code initialization, (c) EHR hypergraph encoding, (d) Knowledge-infused medical text generation, (e) Medical text encoding. The loss functions denoted in the red dashed box. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

(I) Code-level representation learning flow: The process of learning the code-level patient state representation learning involves the following modules:

- **Medical code initialization module**: This module capitalizes on multi-source domain graph structure knowledge related to medical codes. It focuses on learning the initial embeddings of medical codes by encoding the information flow within the graph structure.
- **EHR hypergraph encoding module**: The second module is dedicated to constructing an EHR hypergraph, which is instrumental in further encoding the medical codes with complex hidden relationships, thereby facilitating the acquisition of code-level patient local state representations enriched with structural information.

In the following sections, each module is discussed in detail.

(I.a) Medical code initialization module: In the medication recommendation task, most existing methods (Shang, Xiao et al., 2019; Zhang et al., 2017) typically employ multi-hot encoding for representing medical codes, which often overlooks critical information embedded in medical domain knowledge. Notably, each medication's functionality is largely determined by its molecular

substructures (such as functional groups), with substructure-level graphs offering more predictive insights than atomic-level (Yang, Xiao, Ma et al., 2021) or taxonomic-level criteria.

Furthermore, the diagnosis and procedure codes conform to the principles of ICD coding, which serve as classification standards. The ontology graph, therefore, becomes an indispensable external knowledge base, encapsulating the knowledge inherent in these standards. Within this graph, leaf nodes correspond to individual medical codes, while the ancestor nodes encompass classification information pertinent to medical contexts. Acknowledging the potential of representing the aforementioned external knowledge through graph structures, we propose a general model that encodes graph-structured domain knowledge, facilitating the learning of initial medical code embeddings.

For the medication code:

We utilize the SMILES protocol (Weininger, 1988) to transform the molecular structures of $|M|$ medications into SMILES strings. These are further segmented into $|Sub|$ unique substructures using the BRICS segmentation method (Degen, Wegscheid-Gerlach, Zaliani, & Rarey, 2008). In this approach, each medication is uniquely represented by a substructure graph. This graph consists of nodes, which are a subset of Sub, and edges that represent the connections between these substructures. For each medication, we construct an adjacency matrix $A^{(m)}$, which is a binary matrix with dimensions $|Sub| \times |Sub|$. In this matrix, $A^{(k)}_{i,j|i\neq j} = 1$ indicates a bond between substructures sub_i and sub_j within the kth medication's substructure graph, and $A^{(k)}_{i,j|i\neq j} = 0$ in the absence of a bond. With the initial learnable substructure embedding matrix E_{sub} and the medication substructure adjacency matrix $A^{(m)}$, we employ a specific function to encode the substructure graph:

$$e_{g_i} = F_{\text{Agg}}(g_i, nb(g_i), E_{\text{sub}}), \tag{2}$$

where $F_{\text{Agg}}(\cdot,\cdot,\cdot)$ is an embedding aggregation function, $nb(g_i)$ represents the connect neighbor substructure set to substructure g_i. By applying the embedding aggregation function $F_{\text{Agg}}(\cdot,\cdot,\cdot)$, the substructure embedding e_{g_i} is updated considering the neighboring substructure nodes within the same medication. To derive the final medication embeddings, we use a simple readout function, $\text{avg}(\cdot)$. This function calculates the average of all substructure states linked to a medication, effectively integrating the individual substructure embeddings associated with that medication.

$$m^{(k)} = \text{avg}(\textstyle\sum_{i\in S^{(k)}} e_{g_i}), \tag{3}$$

where $S^{(k)} \subseteq S$ is the substructure set of the kth medication, and $m^{(k)}$ is the resulting medication embedding.

For the diagnosis and procedure code:

The ontology structure establishes hierarchical relationships among medical codes, where parent nodes effectively *summarize* information from child nodes, and conversely, child nodes *inherit* attributes from their parent nodes.

We introduce a general method to model this information flow within the ontology structure. This method involves amalgamating information from child nodes into their respective parent node and subsequently updating the child nodes with information derived from the parent node. For each medical code, the embedding e_c is partitioned into two components: a unique part e^{uni}_c, encapsulating distinct information, and an inherited part e^{inh}_c, representing information acquired from the parent node. The initial code embedding e_c is then constructed by merging these unique and inherited embeddings through a trainable coefficient λ_c,

$$e_c = \lambda_c e^{inh}_c + (1-\lambda_c) e^{uni}_c. \tag{4}$$

Next, we implement a bidirectional information flow (*i.e.*, summarizing and inheriting flow) to facilitate the learning of medical code embeddings. The unique part of a node c_i, denoted as $e^{uni}_{c_i}$, aggregates the unique part embeddings of all its child nodes. Conversely, the inherited part $e^{inh}_{c_j}$ assimilates the complete code embeddings from all parent nodes of c_i. This process is formally represented as:

$$e^{uni}_{c_i} = F_{\text{Agg}}(c_i, ch(c_i), E_{\text{uni}}), \tag{5}$$

$$e^{inh}_{c_i} = F_{\text{Agg}}(c_i, pa(c_i), E_{\text{inh}}), \tag{6}$$

where $ch(c_i)$ and $pa(c_i)$ signify the sets of child and parent nodes, respectively, associated with code c_i. W_{uni} and W_{inh} are the corresponding embedding matrices. The embedding aggregation function facilitates the transmission and combination of information between a target node and its directly connected node set. For the embedding aggregation function, we employ the Graph Attention Network (GAT) (Velickovic et al., 2017), which has proven its efficacy in graph-structured medical embedding tasks in previous studies (Shang, Ma et al., 2019; Wu, He et al., 2023).

By following the two steps outlined earlier, we acquire the embeddings for diagnosis, procedure, and medication codes, denoted as X_d, X_p, and X_m respectively. These are obtained by concatenating the embeddings of all relevant medical codes together, thus forming a comprehensive initialization embedding of medical data,

$$X_c = \text{concat}\left[c^{(1)}, c^{(2)}, \ldots, c^{(|C|)}\right], c \in \{d, p, m\}. \tag{7}$$

(I.b) Structural patient state representation learning: Following the previous module, we acquired initialized medical code embeddings. Traditional methods typically encode medical codes either independently or by considering each code as a node, forming edges between codes from the same visit to learn patient representations. However, these approaches often fall short in effectively capturing the complex interconnection of multiple codes across various patient visits in EHRs.

To address this challenge, PROMISE constructs an EHR hypergraph $\mathcal{H} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = \{v_i\}^{|C|}_{i=1}$ is the node set representing medical codes. $\mathcal{E} = \{e^{(t)}_i\}$ is the hyperedge set representing patient visits. Hyperedges contain an unlimited number of nodes

and provide a versatile method for encoding high-order interactions. However, existing methods for learning hyperedge or hypergraph representations may not be entirely suitable for the medication recommendation task, as they could capture either limited information from the current visit or redundant information across all visits. As underscored in the introduction, a representation focused solely on a single visit might not comprehensively capture a patient's evolving health status over time, thereby necessitating the consideration of time intervals for a more complete assessment. Therefore, PROMISE seeks to encapsulate a thorough representation of a patient's health status, factoring in the temporal connections between health states and incorporating historical visit information.

The process occurs in two steps: first, PROMISE constructs a *Sub-hypergraph* within $\mathcal{H}$ by merging multiple currently and previously visited hyperedges. This ensures that the sub-hypergraph accurately reflects the most recent patient data. The subsequent step involves encoding each sub-hypergraph to represent the patient state.

In each sub-hypergraph, PROMISE employs an attention mechanism with ℓ_s layers to update the embeddings of nodes $v_i \in \mathcal{V}$ and hyperedges $e_j \in \mathcal{E}$. The initial node embedding $v_i^{(0)}$ is derived from the medical code embeddings X_c, while the initial hyperedge embedding $e_j^{(0)}$ is obtained by averaging all $v_i^{(0)}$ associated with $e_j^{(0)}$. For each layer $\ell \in (1, \ell_s)$, the updating process occurs in two phases: firstly, the hyperedge embeddings are updated from $e_j^{(\ell-1)}$ to $e_j^{(\ell)}$; subsequently, the node embeddings are updated from $v_i^{(\ell)}$ to $v_i^{(\ell+1)}$,

$$e_j^{(\ell)} = \sum_{v_i \in e_j} a_E(e_j^{(\ell-1)}, v_i^{(\ell)}) v_i^{(\ell)}, \tag{8}$$

$$v_i^{(\ell+1)} = \sum_{e_j \ni v_i} a_V(v_i^{(\ell)}, e_j^{(\ell)}) e_j^{(\ell)}, \tag{9}$$

where the attention scores $a_E(e_j^{(\ell)}, v_i^{(\ell)})$, $a_V(v_i^{(\ell)}, e_j^{(\ell)})$ follow:

$$a_E(e_j^{(\ell-1)}, v_i^{(\ell)}) = \frac{\exp(s(e_j^{(\ell-1)}, v_i^{(\ell)}))}{\sum_{v_{i'} \in e_j} \exp(s(e_j^{(\ell-1)}, v_{i'}^{(\ell)}))}, \tag{10}$$

$$a_V(v_i^{(\ell)}, e_j^{(\ell)}) = \frac{\exp(s(v_i^{(\ell)}, e_j^{(\ell)}))}{\sum_{e_{j'} \ni v_i} \exp(s(v_i^{(\ell)}, e_{i'}^{(\ell)}))}, \tag{11}$$

$$s(v_i, e_j) = (W_N v_i + b_N) \odot (W_E e_j + b_E), \tag{12}$$

where W_N, W_E, b_N, and b_E are trainable parameters. Due to the superiority of hypergraph, PROMISE leverages the updated hyperedge to enhance the node embedding by incorporating more closely related nodes.

Once the sub-hypergraphs have been updated through ℓ_s layers, PROMISE encodes the sub-hypergraph using a weighted attention mechanism. This mechanism applies attention over both nodes and hyperedges in the following manner:

$$a(\mathcal{G}_j, v_i^{(\ell_s)}) = \frac{\exp({w_v}^T v_i^{(\ell_s)})}{\sum_{v_{i'} \in \mathcal{G}_j} \exp({w_v}^T v_{i'}^{(\ell_s)})}, \tag{13}$$

$$a(\mathcal{G}_j, e_i^{(\ell_s)}) = \frac{\exp({w_e}^T e_i^{(\ell_s)})}{\sum_{e_{i'} \in \mathcal{G}_j} \exp({w_e}^T e_{i'}^{(\ell_s)})}, \tag{14}$$

where w_v and w_e are the learnable parameters, respectively. Utilizing the obtained attention weights, we calculate the code-level patient local state representation, represented as q_C^L. This is achieved by integrating the nodes and hyperedges of each sub-hypergraph in the final layer, denoted as ℓ_s:

$$\mathcal{G}_{j_v} = \sum_{v_i \in \mathcal{G}_j} a(\mathcal{G}_j, v_i^{(\ell_s)}) v_i^{(\ell_s)}, \tag{15}$$

$$\mathcal{G}_{j_e} = \sum_{e_i \in \mathcal{G}_j} a(\mathcal{G}_j, e_i^{(\ell_s)}) e_i^{(\ell_s)}, \tag{16}$$

$$q_C^L = \text{concat}[\mathcal{G}_{j_v}, \mathcal{G}_{j_e}]. \tag{17}$$

This approach ensures a comprehensive representation that encapsulates the intricate relationships and interactions within the sub-hypergraph, reflecting the patient's current health state with high fidelity.

(II) Text-level representation learning flow: The process of learning the text-level patient state representation involves the following modules:

- **Medical text generation module**: This module transforms structured tabular EHR data into an unstructured text format. It achieves this by substituting medical codes with their corresponding formal descriptions derived from domain knowledge, thus facilitating a more natural and comprehensive textual interpretation of the data.
- **Medical text encoding module**: This module utilizes a pre-trained language model to extract text-level patient state representations. It interprets and encodes the semantic information embedded within the unstructured EHR data, encoding the textual content that can be effectively used for further analyzing.

In the following sections, each module is discussed in detail.

(II.a) Unstructured patient clinical record generation: Structured EHR data is typically organized as tabular data and fields in a chronological manner, using standardized coding systems to facilitate analysis. However, the medical codes in EHRs often lack semantic information and are not fully equipped to capture the intricacies of medical events. The transformation of structured EHR into unstructured data involves extracting table fields and values and converting them into free text, thereby enriching the medical records with contextual information about historical visits and the descriptive text of medical codes.

Our methodology initially represents patient tabular data as key–value attribute pairs, inclusive of code information. We then flatten these key–value pairs into code-based unstructured 'sentences' (as illustrated in Fig. 4.(d.1)). Patients' clinical notes are organized into sentences that follow the chronological order of visits, where each sentence corresponds to a specific visit, encapsulating related medical codes and pertinent information. This format offers a more dynamic and comprehensive representation of patient history.

Each medical code is associated with a formal description in its respective classification system, which can be sourced from the official database. Subsequently, as illustrated in Fig. 4.(d.2), we substitute the code information in each sentence with its corresponding formal description. This replacement significantly enhances the semantic interpretation of the medical codes. By following these procedures, we successfully compile the patient's unstructured clinical records, enriching them with detailed and precise medical description information.

(II.b) Semantic patient state representation learning: After obtaining the patient's descriptive clinical records in an unstructured format, we specifically chose to employ the Bio-Clinical BERT (Alsentzer et al., 2019) encoder for deriving semantically rich representations of patient local states. This decision was guided by Bio-Clinical BERT's specialized training on biomedical and clinical corpora, such as the MIMIC-III database (Johnson et al., 2016), which contains extensive EHRs from ICU patients, enabling it to capture the complex semantics in clinical text more effectively than general-domain models or other biomedical embeddings. Incorporating Bio-Clinical BERT into our framework allows us to leverage its domain-specific expertise, ensuring that the semantic information inherent in patient records is accurately captured. This capability is crucial for generating detailed patient state representations, supporting the primary goal of our study to enhance the accuracy and personalization of medication recommendations through a profound understanding of patient data. This process of encoding is visually depicted in Fig. 4.(e), illustrating the transformation of unstructured data into structured, meaningful patient state information.

Preprocessing: In this phase, each input, comprising multiple sentences from unstructured records visited within a specific time interval, is treated as a sequential word array. We incorporate the special symbol [CLS] at the start of each sequence, where the final hidden state of [CLS] symbolizes the entire record's semantic representation.

Encoding: The input, primarily a medical text sequence, includes token embeddings, segment embeddings, and position embeddings. These are amalgamated into an initial embedding matrix. This matrix is then processed through the model's multi-layered Transformer architecture, which encompasses self-attention layers and feed-forward neural networks. We address the Out-Of-Vocabulary (OOV) issue through its tokenization strategy. Specifically, it employs the *WordPiece tokenizer*, which breaks down words into known subwords or tokens. This method allows the model to handle new words that were not explicitly present in the training data but can be constructed from known subwords. Each subword has its own embedding, and the representation of a word that is not in the vocabulary is constructed by combining the embeddings of its subwords. This allows deriving meaningful representations for OOV words based on the embeddings of their constituent parts. The Transformer block's pre-training on extensive EHR corpora, including notes from MIMIC-III and MIMIC-IV, is complemented by subsequent fine-tuning on a corpus of medical codes and descriptions.

Output: We apply an attention pooling mechanism (Er, Zhang, Wang, & Pratama, 2016) to amalgamate the token vectors from the final layer, yielding semantic embeddings. This method effectively captures the entire sequence's semantics, accounting for inter-token relationships.

Through these steps, the Bio-Clinical BERT model transforms the medical clinical records in the time interval into text-level patient local state representation q_T^L.

4.2.3. Multimodal global representation learning

At this stage, our initial task is to identify a cohort of patients exhibiting high similarity to the current patient within each respective learning flow. Following this, we obtain the global patient representation by applying a multi-granularity attention mechanism to these similar patients. This mechanism considers varying levels of detail (*i.e.*, patient-level and trajectory-level), ensuring a comprehensive understanding of patient states. The final step involves utilizing a multimodal enhancement module. This module is designed to discern and integrate interactions between patient representations that have been learned across different modalities, thereby enriching the overall patient model with diverse data insights

(I) Similar patient retrieval module: In clinical practice, accurately measuring patient similarity is pivotal for precise diagnoses and the development of individualized treatment plans. Existing methods generally adhere to two methods: *(1) The traditional method* (Fig. 6(a)) evaluates similarities between the patients raw input (Yin, Liu, Zhang, & Zhang, 2020). However, its major drawback lies in its vulnerability to the influence of irrelevant variables present in the raw inputs. *(2) The global measurement method* (Fig. 6(b)) encodes input into representations for similarity measurement and further outcome prediction (Suo et al., 2017). However, it falls short in its ability to assess similarities at a fine-grained level.

Physicians often assess localized changes in patients' vital signs over various time intervals to determine the local health state. They also consider the overall trend in patient state changes, referred to as the health state trajectory, using similar trajectories from other patients to inform medical decision-making. Drawing inspiration from this clinical practice, our PROMISE framework

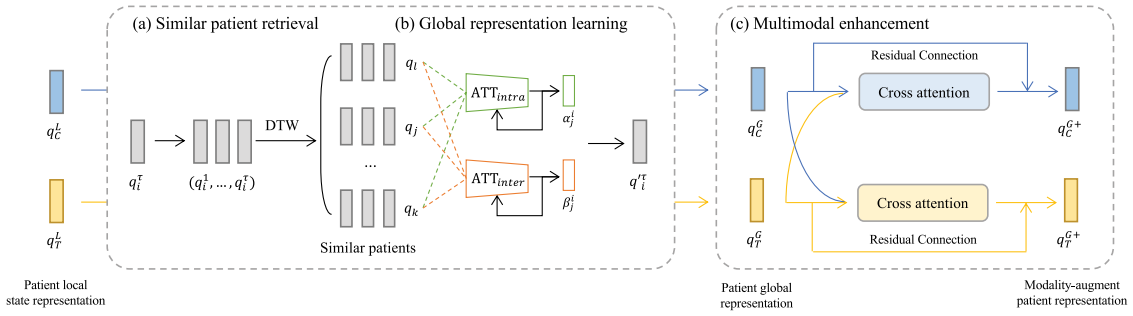


Fig. 5. The architecture of the Multimodal global representation learning stage. The input to this stage is the patient local state representations from code-level and text-level learning flows, and the output of this stage is modality-augment patient representations. PROMISE consists of the following modules in this stage, (a) Similar patient retrieval module, (b) Global representation learning module, and (c) Multimodal enhancement module. The loss functions denoted in the red dashed box. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

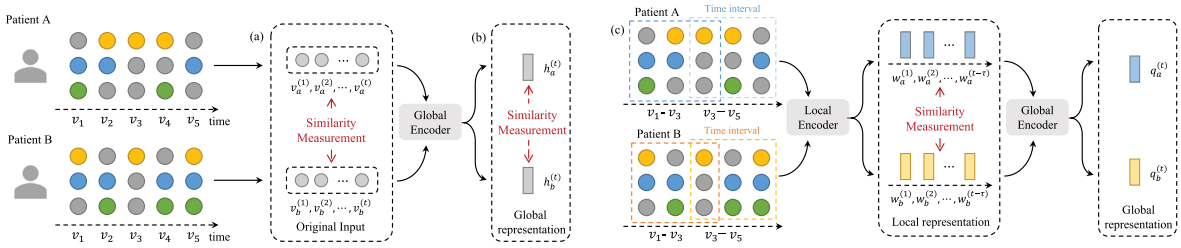


Fig. 6. Differences between existing approaches and our framework. (a) The traditional method directly employs similarity measurement on the patient raw input data. (b) The global measurement method employs learned patient representations through global encoder for similarity measurement and clinical outcome prediction. (c) PROMISE calculates similarity based on health state trajectory representations that synthesize learned local state representations. For outcome prediction, it considers a global representation derived from the aggregate of all state representations.

(Fig. 6(c)) aims to bridge the gap between these two existing methods. It focuses on learning local state representations to capture relevant variables and assess similarities at a detailed level, thereby addressing the limitations inherent in both the primitive and global measurement approaches.

To implement this approach, we start with a time series of patient visits. We dynamically segment these visit records into a series of partially overlapping time intervals. Each time interval is denoted as $T^{k:k+l}$, representing the raw patient health trajectory $V^{k:k+l}$. As outlined in the previous stage 4.2.2, we encoded each time interval into patient state representations across multiple learning flows q_C^L and q_T^L within the time interval $T^{k:k+l}$ (simplified as q^k).

As shown in Fig. 5, we adopt the dynamic time warping (DTW) method (Müller, 2007) to measure the similarity between medical trajectory representations of different patients. For a given patient i, the local state trajectory can be denoted as: $q_i^{\tau_i} = [q_i^{(1)}, q_i^{(2)}, \ldots, q_i^{(\tau_i)}]$. We define a sub-sequence of $q_i^{\tau_i}$ up to a point m as $q_i^m = [q_i^{(1)}, q_i^{(2)}, \ldots, q^{(m)}]$. The distance between the trajectories of patients i and j is calculated as follows:

$$D(q_i, q_j) = \frac{D(q_i^{\tau_i}, q_j^{\tau_j})}{max(|q_i^{\tau_i}|, |q_j^{\tau_j}|)}, \tag{18}$$

$$D(q_i^m, q_j^n) = d(q_i^m, q_j^n) + \min\left\{ D(q_i^{m-1}, q_j^n), D(q_i^m, q_j^{n-1}), D(q_i^{m-1}, q_j^{n-1}) \right\}. \tag{19}$$

The Euclidean distance, defined as $d(q_i^m, q_j^n) = \|q_i^m - q_j^n\|$, is utilized for analyzing medical trajectories due to its widespread applicability, computational efficiency, and suitability for the specific demands of medical data. Euclidean distance is straightforward and intuitive, making it ideal for quantifying direct distances in time-series health data. This metric excels in scenarios with complex or large datasets due to its low computational demands, ensuring efficiency. Furthermore, while other metrics like Manhattan or cosine similarity have their uses, Euclidean distance is generally more appropriate for medical trajectory analysis where accurate distance measurements are crucial. This choice is also reinforced by its ability to reflect the true spatial relationships in medical data, essential for accurately tracking disease progression. The boundary conditions are given by:

$$D(q_i^1, q_j^n) = \sum_{k=1}^{n} d(q_i^1, q_j^k), \quad D(q_i^m, q_j^1) = \sum_{k=1}^{m} d(q_i^k, q_j^1). \tag{20}$$

We then use $D(q_i, q_j)$ as the cumulative distance along the matching path, where a smaller distance score indicates higher similarity. To maintain computational efficiency, we limit our analysis to the top n patient state trajectories. The top n similar patient trajectories are denoted as: $q_k^{1:\tau_k} = \{q_k^{(1)}, \ldots, q_k^{(\tau_k)}\}$, where k ranges from 1 to n.

(II) Global representation learning module: In the previous module, we identified the top n patient trajectories $q_k^{1:\tau_k}, k \in (1,\ldots,n)$ that exhibit high similarity to the target patient trajectory $q_i^{1:\tau_i}$. In this module, our objective is to learn the global patient representation from these highly similar patient trajectories by employing two distinct attention mechanisms: *trajectory-level attention* and *patient-level attention*. These mechanisms assign varying weights to different medical time intervals and patient trajectories, respectively.

Firstly, the *trajectory-level attention* focuses on pinpointing crucial time intervals within each trajectory. This involves assigning trajectory-level weights $\{\alpha_i^{(1)},\ldots,\alpha_i^{(k)},\ldots,\alpha_i^{(n)}\}, k \in ((1,n),i)$ to time intervals in each patient trajectory $\{q_1^{1:\tau_1},\ldots,q_k^{1:\tau_k},\ldots,q_n^{1:\tau_n}\}, k \in ((1,n),i)$ as follows:

$$g_i^{(k)} = \mathrm{RNN_T}(g_i^{(k-1)}, q_k^{1:\tau_k}), \tag{21}$$

$$\alpha_i^{(k)} = \tanh(W_T g_i^{(k)} + b_T), \tag{22}$$

where W_T, b_T are learnable parameters.

Next, the *patient-level attention* is employed to determine the relative importance of patient within the similar patients set by assigning weights $\{\beta_i^{(1)},\ldots,\beta_i^{(k)},\ldots,\beta_i^{(n)}\}, k \in ((1,n),i)$ to each patient trajectory $\{q_1^{1:\tau_1},\ldots,q_k^{1:\tau_k},\ldots,q_n^{1:\tau_n}\}, k \in ((1,n),i)$ as follows:

$$h_i^{(k)} = \mathrm{RNN_P}(h_i^{(k-1)}, q_k^{1:\tau_k}), \tag{23}$$

$$\beta_i^{(k)} = \frac{W_P h_i^{(k)}}{\sum_{k=1}^{n} W_P h_i^{(k)}}, \tag{24}$$

where W_P is a learnable parameter.

By integrating these two attention mechanisms with the health trajectories of multiple patients, we assign the highest weight to the most relevant patient trajectory and critical time intervals within the health trajectories. This allows us to derive the global patient representation from the local patient representations, denoted as q_i^G.

$$q_i^G = q'^{\tau_i}_i = \sum_{k=1}^{n,i} \beta_i^{(k)} \cdot (\alpha_i^{(k)} \odot q_k^{1:\tau_k}), \tag{25}$$

where $q_k^{1:\tau_k}, k \in ((1,n),i)$ denotes the health trajectories of patient i and top n similar patients, $\odot$ is the element-wise multiplication. In this way, we obtain the multimodal patient global representations q_C^G and q_T^G in the code-level and text-level learning flow, respectively.

(III) Multimodal enhancement module: In this module, we incorporate a cross-attention mechanism to capture correlations between multimodal patient representations from different learning flows, while preserving modality-specific information through residual connections. Specifically, we utilize the code-level patient global representation as a query to extract pertinent semantic information from the text-level patient representation. Simultaneously, a similar attention-based query is conducted on the text-level patient global representation to gather relevant structural information from the code-level patient representation. The cross-attention on the two modalities is denoted as:

$$q_C^{G+} = \mathrm{Softmax}\left(q_T^G \cdot q_C^{G\mathrm{T}} / \sqrt{d}\right) \cdot q_C^G + q_T^G, \tag{26}$$

$$q_T^{G+} = \mathrm{Softmax}\left(q_C^G \cdot q_T^{G\mathrm{T}} / \sqrt{d}\right) \cdot q_T^G + q_C^G, \tag{27}$$

where q_C^{G+} and q_T^{G+} represent the modality-augment code-level and text-level patient representation at the current visit t. For convenience, we simplify it as $q_C^{(t)}$ and $q_T^{(t)}$.

The augmented multimodal patient representation is then combined to form the comprehensive patient representation for the current visit, denoted as $h^{(t)}$. PROMISE then proceeds to provide medication recommendations based on the patient representation $h^{(t)}$.

$$h^{(t)} = \mathrm{concat}\left[q_C^{(t)}; q_T^{(t)}\right], \tag{28}$$

$$o^{(t)} = \mathrm{Softmax}(W_m h^{(t)} + b_m), \tag{29}$$

where W_m and b_m are learnable parameters during the training process. $o^{(t)}$ signifies the final predicted probability for a patient at a specific time t. By comparing this predicted probability $o^{(t)}$ with a pre-defined threshold parameter δ, the final medication recommendation, denoted as $\hat{o}^{(t)}$, can be determined, which is crucial for transforming the probabilistic outputs of the framework into actionable medication combinations.

4.2.4. Pre-training and fine-tuning

To achieve the effective patient representation for the medication recommendation, our PROMISE includes the following pre-training and fine-tuning stage (see Fig. 7).

(I) The pre-training stage: PROMISE undergoes a pre-training process utilizing multiple loss functions to optimize its performance:

- **Medical code pre-training**: This phase focuses on constraining and acquiring the initial embeddings for medical codes, leveraging domain-specific knowledge. The process involves a general loss function, augmented with distance constraints, to ensure that the embeddings accurately reflect the inherent structure and relationships within the medical codes.
- **Information constrained pre-training**: This stage is dedicated to discerning correlations between multimodal patient representations. It applies information constraints across different modalities and within each modality-specific learning flow. Specifically, it aims to reveal intricate relationships among patient representations from diverse sources and further refines the representations within each modality.

In the following sections, each pre-training task is discussed in detail.

The medical code pre-training

During the pre-training phase for diagnosis and procedure medical codes, we incorporate their respective ontology structures as a form of external knowledge. This approach leverages the hierarchical and relational information inherent in these ontologies to enrich the understanding and representation of the medical codes. The primary objective in this phase is to minimize the concept divergence of medical code embeddings. To achieve this goal, our approach concentrates on minimizing the distance between connected nodes, while at the same time, maximizing the distance between unconnected nodes within the ontology structure, which is crucial in capturing the inherent relationships and hierarchies present in the medical codes. The adjacency matrix $\mathcal{G}^o$ is employed to represent the set of connected nodes within the medical ontology, effectively illustrating the connections and associations among different nodes in the structure.

For medication codes, we incorporate knowledge of drug-drug interactions (DDIs) as an external source of information, particularly focusing on two main types: synergistic DDIs (SDDIs) and antagonistic DDIs (ADDIs). In cases where a pair of medications demonstrates SDDIs, administering them concurrently could potentially increase their collective efficacy in treating the relevant illness. On the other hand, if a medication combination exhibits ADDIs, this may result in unwanted side effects or reduce the effectiveness of the individual components. This consideration of DDIs is crucial in optimizing medication recommendations and ensuring patient safety. Accordingly, our approach aims to minimize the distance between SDDI pairs while maximizing the distance between ADDI pairs, thereby augmenting the safety and efficacy of medication recommendations. To represent the relationships between medications that exhibit synergistic DDIs (SDDIs) and antagonistic DDIs (ADDIs), we employ two distinct adjacency matrices: $\mathcal{G}^s$ for SDDIs and $\mathcal{G}^a$ for ADDIs. These matrices provide a clear and structured representation of these critical drug-drug interactions.

To enforce the distance constraint, we initially define the hyperbolic distance (Krioukov, Papadopoulos, Kitsak, Vahdat, & Boguná, 2010) $d(e_i, e_j)$ between two embeddings e_i and e_j as follows:

$$d(e_i, e_j) = \cosh^{-1}\left(1 + 2 \times \frac{\|e_i - e_j\|^2}{(1 - \|e_i\|^2)(1 - \|e_j\|^2)}\right). \tag{30}$$

Secondly, we form triplets in the format (e_i, e_j, e_k). Here, e_i, e_j, and e_k represent embeddings of different entities. The distance-constrained loss function, which incorporates a margin θ, is defined to achieve two objectives: firstly, to minimize the distance between the pair (e_i, e_j), and secondly, to maximize the distance between the pair (e_i, e_k). The distance-constrained loss function can be formulated as:

$$\mathcal{L}_{\text{dist}} = -\sum_{i,j \in \mathcal{G}} \log\left(\frac{e^{-d(e_i, e_j)}}{\sum_{k \in \mathcal{G}'} e^{-d(e_i, e_k)} + \theta}\right). \tag{31}$$

In the context of the medical ontology for diagnosis and procedure, the pair (e_i, e_j) signifies the nodes of connected codes within $\mathcal{G}^o$, representing a direct relationship. Conversely, the pair (e_i, e_k) denotes nodes of non-connected codes, indicating a lack of direct relationship. For the medication codes, the interpretation differs slightly. (e_i, e_j, e_k) corresponds to the ith, jth, and kth medications. Here, (e_i, e_j) represents two medications that form a synergistic pair, as indicated in $\mathcal{G}^s$. This suggests that these medications, when used together, may enhance each other's effectiveness. On the other hand, (e_i, e_k) represents two medications that form an antagonistic pair, as identified in $\mathcal{G}^a$, implying potential adverse interactions or reduced effectiveness when combined.

Here, we obtain the final initialized embeddings for diagnosis, procedure, and medication, denoted as X_d, X_p, and X_m, through this medical code pre-training process. These embeddings are then effectively utilized in the EHR hypergraph encoding module, ensuring that our framework is informed by rich, structured representations of medical codes, enhancing its predictive accuracy and relevance.

The information constrained pre-training

To further explore the associations across multiple modalities and to learn robust patient representations within different learning flows, we introduce two optimization objectives: *modality-enhanced learning* and *modality-specific prediction*. In *modality-enhanced learning*, the primary objective is to enhance the model's capacity to effectively integrate and utilize information across different modalities. This is achieved by increasing the mutual information between the representations of these modalities, subsequently leading to improved overall model performance. The *modality-specific prediction* aims to improve the model's performance by reducing the distance between representations of various modalities, once mapped onto a unified feature space, and aligning them closely with the ground truth. This approach is designed to enhance the model's performance in accurately predicting outcomes based on multimodal data. These two optimization objectives are based on modality-augmented patient representations $q_C^{(t)}$ and $q_T^{(t)}$.

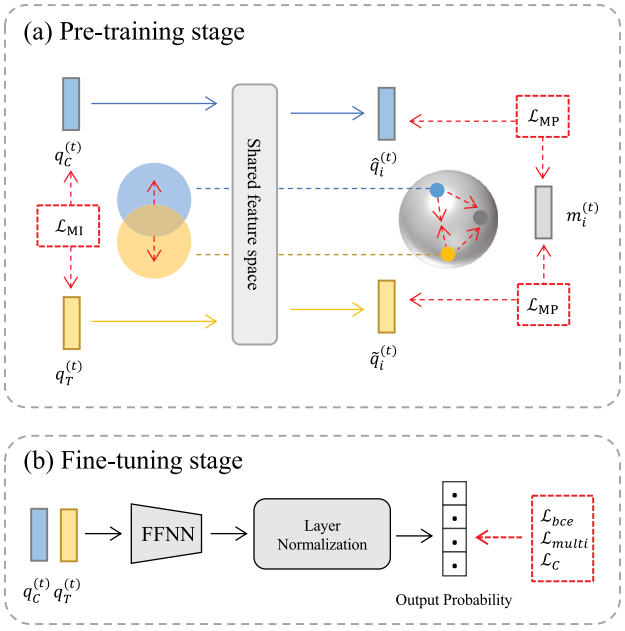


Fig. 7. The architecture of the (a) pre-training stage, and (b) fine-tuning stage. The input to this stage is modality-augment patient representations in both modalities, and the output of the fine-tuning stage is the medication prediction probabilities. The loss functions denoted in the red dashed box. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

In the *modality-enhanced learning*, the goal is to increase the mutual information between modality-enhanced patient representations. This approach serves two purposes: firstly, it compensates for the information gaps in any one modality by leveraging data from others, thereby enhancing the model's robustness. Secondly, it encourages the model to integrate disparate modal information effectively, facilitating the learning of more generalized features and ensuring consistent representations across different modalities.

To realize this objective, we focus on optimizing the lower bound of the mutual information $\text{MI}(q_C^{(t)}, q_T^{(t)})$. We utilize an enhanced InfoNCE loss function that is specifically designed to accommodates the symmetry inherent in multimodal interactions. Within each batch $\mathcal{B}$, a sample from each modality is treated equivalently as an anchor. The representations of two modalities from the same patient are coupled to form positive samples, and the representations of randomly selected patients in the training batch are paired as negative samples. This pairing approach ensures that positive samples reflect congruent multimodal data from a single patient, while negative samples represent disparate data from distinct patients, aiding in effectively training the model to distinguish relevant from irrelevant multimodal relationships. The enhanced InfoNCE loss is formulated as follows:

$$\mathcal{L}_{\text{MI}} = -\frac{1}{2\mathcal{B}} \sum_{i=1}^{\mathcal{B}} [\log \frac{f(q_C^{(t)}, q_T^{(t)})}{\sum_{i=1}^{\mathcal{B}} f(q_C^{(t)}, q_{T_i}^{(t)})} + \log \frac{f(q_T^{(t)}, q_C^{(t)})}{\sum_{i=1}^{\mathcal{B}} f(q_T^{(t)}, q_{C_i}^{(t)})}], \tag{32}$$

where, $q_C^{(t)}$ and $q_T^{(t)}$ represent the anchor and positive sample from code and text modalities, $q_{T_i}^{(t)}$ and $q_{C_i}^{(t)}$ are the negative samples from two modalities, respectively. $f(\cdot,\cdot) = \exp(\text{sim}(\cdot,\cdot)/\tau)$ applies the dot product similarity measurement with a temperature parameter τ. By minimizing this bidirectional InfoNCE loss, the model learns representations that capture shared information between the modalities $q_C^{(t)}$ and $q_T^{(t)}$.

Maximizing mutual information in multimodal representation learning plays an important role in reducing redundancy, yielding more integrated, generalizable, and robust representations. This process facilitates the efficient convergence of various modalities into a shared feature space, streamlining the process of interaction and integration of information within the multimodal framework.

The *modality-specific prediction* pre-training task seeks to enhance the model's performance by reducing the distance between representations of different modalities, once these are mapped onto a shared feature space, and aligning them closely with the ground truth. This strategy enhances the capability for effective comparison and analysis of data across different modalities, offering a more unified and comprehensive analytical perspective. We convert the modality-enhanced patient representations $q_C^{(t)}$ and $q_T^{(t)}$ into a shared feature space.

$$\begin{aligned} \hat{q}_i^{(t)} &= q_C^{(t)} W_s + b_s, \\ \tilde{q}_i^{(t)} &= q_T^{(t)} W_s + b_s, \end{aligned} \tag{33}$$

where, W_s, b_s are learnable parameters in shared feature space. $\hat{q}_i^{(t)}$ and $\tilde{q}_i^{(t)}$ are the mapping of $q_C^{(t)}$ and $q_T^{(t)}$ in the shared feature space. Next, we pair $\hat{q}_i^{(t)}$ and $\tilde{q}_i^{(t)}$ with the label $m_i^{(t)}$ of the target medications as a positive sample. Additionally, in our training

process, we pair each medication with labels of other medications from the same batch, $\mathcal{B}$, as negative samples. The objective of this approach is to use a loss function that enhances the similarity between positive pairs while minimizing the similarity between negative pairs. The loss function can be expressed as follows:

$$\mathcal{L}_{\text{MP}} = -\sum_{i=1}^{|\mathcal{B}|} \log \frac{f(\hat{q}_i^{(t)}, m_i^{(t)}) + f(\tilde{q}_i^{(t)}, m_i^{(t)})}{\sum_{i'=1}^{|\mathcal{B}|} [f(\hat{q}_i^{(t)}, m_{i'}^{(t)}) + f(\tilde{q}_i^{(t)}, m_{i'}^{(t)})]}, \tag{34}$$

where $f(\cdot,\cdot) = \exp(\text{sim}(\cdot,\cdot)/\tau)$ applies the similarity measurement with a temperature parameter τ.

During the training stage, we first utilize the *medical code pre-training* functions to pre-train the code initialization module. Following this, we freeze the parameters of the medical code initialization module and concentrate on training the parameters of the multimodal local state representation and global patient representation learning stage. This training process involves minimizing a combined loss, which can be formulated as follows:

$$\mathcal{L}_{\text{pre}} = (1-\mu_{pre})\mathcal{L}_{\text{MP}} + \mu_{pre}\mathcal{L}_{\text{MI}}, \tag{35}$$

where μ_{pre} is a hyper-parameter employed to maintain a balance between these two loss functions.

(II) The fine-tuning stage: In the fine-tuning stage, PROMISE is trained with the multiple loss functions: (1) the *Multi-label prediction loss* to ensure accurate medication recommendations, and (2) the *Controllable DDI loss* to enhance the safety and effectiveness of medication recommendations.

(a) Multi-label prediction loss

Following the methodologies established in previous works (Shang, Xiao et al., 2019; Yang, Xiao, Ma et al., 2021), which treat medication recommendation as a multi-label classification task. To effectively train our framework, we employ two multi-label loss functions.

The first loss function is the *Multi-label margin* loss (Ji & Ye, 2009). This function aims to guarantee that the predicted probability of the true labels surpasses that of the false labels by a specific margin. The formulation of this loss function is as follows:

$$\mathcal{L}_{\text{multi}} = \sum_{i,j:m_i^{(t)}=1, m_j^{(t)}=0} \frac{\max(0, 1-(o_i^{(t)} - o_j^{(t)}))}{|M|}. \tag{36}$$

The second loss function is the *Binary Cross-Entropy* loss. This loss function is particularly suitable for multi-label classification tasks, as it treats each label of the dataset independently. The formulation of this loss function is as follows:

$$\mathcal{L}_{\text{bce}} = -\sum_{i=1}^{|M|} [m_i{}^{(t)} \log(o_i{}^{(t)}) + (1-m_i{}^{(t)}) \log(1-o_i{}^{(t)})]. \tag{37}$$

The multi-label prediction loss, denoted as $\mathcal{L}_{\text{mp}}$, is derived by integrating the aforementioned two loss functions. This combination is adjusted using a balance hyper-parameter μ_{mp}. The formulation of $\mathcal{L}_{\text{mp}}$ can be expressed as follows:

$$\mathcal{L}_{\text{mp}} = \mu_{mp}\mathcal{L}_{\text{multi}} + (1-\mu_{mp})\mathcal{L}_{\text{bce}}. \tag{38}$$

(b) Controllable DDI loss

The importance of balancing accuracy and safety in medication recommendations has been underscored in previous research (Yang, Xiao, Ma et al., 2021; Yang et al., 2023). In our framework, we aim to extend this focus to not only include the safety of medication combinations but also their effectiveness. To achieve this goal, we introduce the Controllable DDI loss, denoted as $\mathcal{L}_{\text{c}}$.

This loss function is specifically designed to minimize the ADDI rate, thereby reducing the likelihood of potential adverse side effects. Concurrently, it aims to maximize the SDDI rate, thereby enhancing the overall effectiveness of the medication combination. The SDDI and ADDI rates can be formulated as follows:

$$\begin{aligned}
\text{SDDI}^{(t)} &= \frac{\sum_{k,l\in\{i:\hat{\mathbf{o}}_i^{(t)}=1\}} \mathbf{1}\{\mathcal{G}_{kl}^s = 1\}}{\sum_{k,l\in\{i:\hat{\mathbf{o}}_i^{(t)}=1\}} 1}, \\
\text{ADDI}^{(t)} &= \frac{\sum_{k,l\in\{i:\hat{\mathbf{o}}_i^{(t)}=1\}} \mathbf{1}\{\mathcal{G}_{kl}^a = 1\}}{\sum_{k,l\in\{i:\hat{\mathbf{o}}_i^{(t)}=1\}} 1}.
\end{aligned} \tag{39}$$

To stabilize and facilitate the training process, we propose an enhanced weight annealing scheme. We begin by pre-defining target rates for both SDDIs and ADDIs, denoted as γ_s and γ_a respectively. These target rates serve as benchmarks against which the actual predicted DDI rates are compared. The deviations between the predicted DDIs (both SDDIs and ADDIs) and their respective target rates are then calculated. These deviations are referred to as DDI margins, denoted as ψ_a for the ADDIs and ψ_s for the SDDIs. These margins essentially quantify the extent to which the current predictions differ from the desired target rates, and adjusting the model to minimize these margins helps align the predicted drug interactions more closely with the defined safety and efficacy targets.

$$\psi_s^{(t)} = \max(0, \gamma_s / \mathrm{SDDI}^{(t)}),$$
$$\psi_a^{(t)} = \max(0, \mathrm{ADDI}^{(t)} / \gamma_a). \tag{40}$$

Now, the controllable DDI loss function $\mathcal{L}_{\mathrm{c}}$ can be formulated according to the DDI margins $\psi_s^{(t)}$, $\psi_a^{(t)}$ as :

$$\begin{aligned}\mathcal{L}_{\mathrm{c}} &= \frac{\psi_s^{(t)}}{\psi_s^{(t)} + \psi_a^{(t)}} \mathcal{L}_{\mathrm{s}} + \frac{\psi_a^{(t)}}{\psi_s^{(t)} + \psi_a^{(t)}} \mathcal{L}_{\mathrm{a}}, \\ &= \frac{\psi_s^{(t)}}{\psi_s^{(t)} + \psi_a^{(t)}} (1 - \sum_{i=1}^{|M|} \sum_{j=1}^{|M|} \mathcal{G}_{i,j}^{s} \cdot \hat{o}_i^{(t)} \cdot \hat{o}_j^{(t)}) + \frac{\psi_a^{(t)}}{\psi_s^{(t)} + \psi_a^{(t)}} (\sum_{i=1}^{|M|} \sum_{j=1}^{|M|} \mathcal{G}_{i,j}^{a} \cdot \hat{o}_i^{(t)} \cdot \hat{o}_j^{(t)}).\end{aligned} \tag{41}$$

Through this method, we dynamically adjust the coefficients for the SDDI loss ($\mathcal{L}_s$) and ADDI loss ($\mathcal{L}_a$) during training.

(c) Total loss function The combined fine-tuning loss function can be formulated as:

$$\mathcal{L}_{\mathrm{mr}} = \mu_c \mathcal{L}_{\mathrm{mp}} + (1 - \mu_c) \mathcal{L}_{\mathrm{c}}, \tag{42}$$

where μ_c is for further balancing the multi-label prediction loss $\mathcal{L}_{mp}$ and the controllable DDI loss function $\mathcal{L}_c$. While $\mathrm{SDDI}^{(t)} \geq \gamma_s$ and $\mathrm{ADDI}^{(t)} \leq \gamma_a$, $\mu_c = 1$, for the other situations,

$$\mu_c = \begin{cases} 1, & \mathrm{SDDI}^{(t)} \geq \gamma_s, \mathrm{ADDI}^{(t)} \leq \gamma_a \\ \min\{1, \exp(\tau(1 - (\psi_s^{(t)} + \psi_s^{(t)})))\}, & others \end{cases}, \tag{43}$$

where τ is a hyper-parameter. When there is a large difference between the actual DDI rate and the target, the value of μ_c slowly decreases to keep the controllable DDI loss weighted heavily. As the gap narrows, μ_c drops quickly, leading the model to focus more on improving the multi-label prediction loss.

4.3. Experiment setup

4.3.1. Dataset

Our experiments utilized the widely accessible two databases: (1) MIMIC-III Database (Johnson et al., 2016): Accessible via source link,[1] this dataset offers comprehensive clinical data from the ICU departments, crucial for various medical research projects. (2) MIMIC-IV Database (Johnson et al., 2023): Accessible via source link,[2] providing enriched clinical data for advanced research purposes. Both datasets are governed under the *PhysioNet Credentialed Health Data License 1.5.0*[3] and require adherence to the *PhysioNet Credentialed Health Data Use Agreement 1.5.0*[4] to ensure the ethical use of the data (Goldberger et al., 2000). These resources are instrumental in advancing medical research by offering a real-world clinical data foundation for analysis, modeling, and various studies aimed at improving patient care and outcomes.

Our data processing for these datasets involved compiling diagnosis, procedure, and medication records from each patient visit, using various clinical tables (Yang, Xiao, Ma et al., 2021; Yang et al., 2023). To ensure comprehensive historical health information for our analyses, we included only patients with multiple visits and considered medication records within the first 24 h of each visit.

For encoding diagnosis and procedure codes, we used the ICD-9 system. Medication embeddings were based on the ATC-3 classification and molecular substructures from DrugBank. We focused on the top 40 most prevalent drug-drug interactions (DDIs) out of the 86 listed in DrugBank (Wishart et al., 2018), representing medications by their DrugBank and PubChem IDs and integrating this data by mapping to the ATC 3rd level. DDIs were classified into synergistic, antagonistic, and no interaction categories with expert guidance. Table 2 details our dataset specifics. Table 2 provides a detailed breakdown of dataset's specifics.[5]

4.3.2. Baselines

We assess the performance of the PROMISE framework by comparing it to the following ten baseline methods. These methods were chosen to cover a broad spectrum of methodologies in medication recommendation models, ranging from traditional statistical methods to advanced deep learning techniques. The selection of the ten approaches, all of which have shown excellent performance in recent years, and they are all tested on the MIMIC-III dataset. By employing these methods, tested on the same dataset, we ensure a consistent basis for evaluating our framework's performance. This selection allows us to benchmark our proposed framework against a diverse set of models, highlighting its effectiveness and efficiency.

The baseline methods including Logistic Regression (LR), Ensembled Classifier Chain (ECC) (Read, Pfahringer, Holmes, & Frank, 2011), LEAP (Zhang et al., 2017), RETAIN (Choi, Bahadori, Sun et al., 2016), GAMENet (Shang, Xiao et al., 2019), SafeDrug (Yang, Xiao, Ma et al., 2021), MICRON (Yang et al., 2021), COGNet (Wu, Qiu et al., 2022), DrugRec (Sun et al., 2022), 4SDrug (Tan et al., 2022), and MoleRec (Yang et al., 2023). These baseline methods are categorized as follows: (1) Instance-based methods: LR, ECC, and LEAP. (2) Longitudinal-based methods: RETAIN, GAMENet, SafeDrug, MICRON, COGNet, DrugRec, 4SDrug, MoleRec. To be fair, we tune the hyper-parameters of each model to achieve the best results.

[1] https://physionet.org/content/mimiciii/1.4/.
[2] https://physionet.org/content/mimiciv/2.2/.
[3] https://physionet.org/content/mimiciv/view-license/2.2/.
[4] https://physionet.org/content/mimiciii/view-dua/1.4/.
[5] The preprocessing code for these datasets and the baselines were made publicly.

Table 2
The datasets statistics for MIMIC-III and MIMIC-IV.

Statistics	MIMIC-III	MIMIC-IV
Num. of patients	6350	64,519
Num. of clinical events	15,032	172,400
Num. of diagnoses	1958	2000
Num. of procedures	1430	11,259
Num. of medications	132	128
Num. of visits per patient (avg./max)	2.37/29	2.68/70
Num. of diagnoses per visit (avg./max)	10.51/128	8.24/270
Num. of procedures per visit (avg./max)	3.84/50	2.37/95
Num. of medications per visit (avg./max)	11.43/64	6.74/72
Inherent ADDI ground truth	0.0813	0.0815
Inherent SDDI ground truth	0.1583	0.1665

- **Logistic Regression (LR)** (Indra, Wikarsa, & Turang, 2016): A traditional recommendation technique which utilizes instance-based linear classifier with $L2$ regularization.
- **Ensemble Classifier Chain (ECC)** (Read et al., 2011): A multi-label model using a sequence of classifiers. Each classifier in the chain considers the predictions of previous classifiers as additional inputs.
- **LEAP** (Zhang et al., 2017): An instance-based approach formulates medication recommendation as a sequential decision problem, utilizes a recurrent decoder to model label dependencies, and content-based attention to link labels with instances.
- **RETAIN** (Choi, Bahadori, Sun et al., 2016): Offers sequence prediction via a two-level RNN with neural attention, identifying key past visits and significant clinical variables.
- **4SDrug** (Tan et al., 2022): Uses set-oriented representations, similarity measurements, and importance-based set aggregation for symptoms, alongside intersection-based set augmentation for medication recommendations.
- **GAMENet** (Shang, Xiao et al., 2019): Incorporates memory-augmented neural networks with a memory bank that integrates medication usage history and a Drug-Drug Interaction graph.
- **SafeDrug** (Yang, Xiao, Ma et al., 2021): Utilizes both global and local encoders to capture molecular structure information, enhancing the medication recommendation process.
- **MICRON** (Yang et al., 2021): Introduces a recurrent residual learning model to link drug changes and disease changes and focus on incremental learning of predicting changes in medication.
- **COGNet** (Wu, Qiu et al., 2022): Employs a copy-or-predict mechanism and uses Transformer for generative inference, generating medication combinations based on a patient's past medication history.
- **DrugRec** (Sun et al., 2022): A causal inference-based medication recommendation model that addresses recommendation bias. It leverages historical health conditions and coordinates multiple medications for safety.
- **MoleRec** (Yang et al., 2023): Utilizes a hierarchical architecture to model interactions between substructures and the influence of individual substructures on a patient's health condition, specifically for medication recommendation.

4.3.3. Evaluation metrics

To assess the performance of PROMISE in terms of accuracy, efficacy, and safety in medication recommendations, various metrics are utilized. These include:

- **SDDI Rate**: Measures the effectiveness of medication combinations.
- **ADDI Rate**: Assesses the safety by quantifying potential adverse interactions.

The calculation methods for both SDDI and ADDI rates are detailed in . Additionally, PROMISE employs the following metrics:

- **Jaccard Similarity Score (Jaccard)**: Jaccard is a statistical measure of the similarity and diversity between sample sets (Niwattanakul, Singthongchai, Naenudorn, & Wanapu, 2013).

$$\text{Jaccard}_i = \frac{1}{T_i}\sum_{t=1}^{T_i}\frac{|\{o_{i,j}^{(t)}=1\}\cap\{\hat{o}_{i,j}^{(t)}=1\}|}{|\{o_{i,j}^{(t)}=1\}\cup\{\hat{o}_{i,j}^{(t)}=1\}|}. \tag{44}$$

- **Precision-Recall Area Under the Curve (PRAUC)**: PRAUC evaluates the trade-off between precision and recall for different threshold settings (Davis & Goadrich, 2006).

$$\text{PRAUC}_i = \frac{1}{T_i}\sum_{t=1}^{T_i}\sum_{k=1}^{|M|}\text{P}(k)_i^{(t)}(\text{R}(k)_i^{(t)}-\text{R}(k-1)_i^{(t)}), \tag{45}$$

where k represents the rank within the sequence of retrieved medications.

- **Average F1 Score (F1)**: F1 harmonic mean of precision and recall, providing a balance between them.

$$\text{F1}_i = \frac{1}{T_i}\sum_{t=1}^{T_i}\frac{2\text{P}_i^{(t)}\text{R}_i^{(t)}}{\text{P}_i^{(t)}+\text{R}_i^{(t)}}, \tag{46}$$

where we provide the definitions of Precision $P_i^{(t)}$ and Recall $R_i^{(t)}$ for a patient i at the tth visit,

$$P_i^{(t)} = \frac{|\{m_{i,j}^{(t)} = 1\} \cap \{\hat{o}_{i,j}^{(t)} = 1\}|}{|\{\hat{o}_{i,j}^{(t)} = 1t\}|}, \tag{47}$$

$$R_i^{(t)} = \frac{|\{m_{i,j}^{(t)} = 1\} \cap \{\hat{o}_{i,j}^{(t)} = 1\}|}{\{m_{i,j}^{(t)} = 1\}|}. \tag{48}$$

All these metrics are macro-averaged to provide a comprehensive evaluation across various dimensions.

4.3.4. Implementation details

For our medication recommendation task, we followed a data splitting strategy similar to that used in prior studies (Shang, Xiao et al., 2019; Wu, Qiu et al., 2022; Yang, Xiao, Ma et al., 2021). The dataset was divided into pre-training, fine-tuning, and testing sets in a $\frac{2}{3} : \frac{1}{6} : \frac{1}{6}$ ratio, based on patient IDs, to further prevent information leakage. This split ensures ample data for pre-training while keeping distinct sets for fine-tuning and testing. During pre-training iterations, the fine-tuning set was used to identify optimal model parameters, aiding in model refinement and selection of the best parameter configurations. During our evaluation phase, we used a bootstrapping sampling technique, as suggested in Yang et al. (2021). This involved training models on a set training set and choosing hyperparameters based on a fixed validation set. In each of 10 evaluation rounds, we randomly sampled 80% of the test set data with replacement. The mean and standard deviation from these rounds were reported as the final results, offering a robust assessment of the model's performance. Due to the random nature of sampling, slight variations in the final results are expected.

In our framework, the dimensions of E_d, E_p, E_m, the size of hidden layers in the transformer, and the batch size were all set to 64. The threshold δ for predicting medications $o^{(t)}$ was set to 0.5, serving as the decision boundary for medication recommendations. A dropout rate of 0.6 is selected for our neural network layers. For the semantic learning module, the embedding layer of the pretrained Bio-Clinical BERT remains fixed, with its parameters preserved as originally configured in the fine-tuning phase. These approaches are adopted to mitigate the risk of overfitting. For the medical code initialization module, the number of hidden layers ℓ in the Graph Attention Network (GAT) was set to 2. The weight μ_{pre} in the information-constrained pre-training loss $\mathcal{L}_{pre}$ was 0.1, while the weight μ_{mp} in the multi-label prediction loss $\mathcal{L}_{mp}$ was 0.75. The controlling parameters γ_a and γ_s in the controllable DDI loss $\mathcal{L}_c$ were set to 0.07 and 0.17, respectively. The SafeDrug used γ_a as 0.06 in the original paper, for a fair comparison, we set the γ_a to all baseline methods as 0.07. All models were optimized using the Adam optimizer (Kingma & Ba, 2014), with a learning rate of 1×10^{-4} over 50 epochs. Our training process was executed using 4 NVIDIA V100 GPUs. The baseline models were trained and implemented with optimized parameters, as specified in the referenced studies.

5. Result and discussion

5.1. Main results

Table 3 and Fig. 8 present the results of the performance comparison across different datasets. The efficacy of medication recommendation is assessed using Jaccard, F1 and PRAUC scores, as depicted in Fig. 8(a–c), respectively. In the presentation of the results, the symbol ↑ is used to indicate that a higher value of the corresponding metric represents a better outcome. Conversely, the symbol ↓ is used to signify that a lower value of the corresponding metric is more desirable. In these metrics, higher scores correlate with improved predictive accuracy of the models. Furthermore, Fig. 8(d) illustrates the ADDI Rate, and Fig. 8(e) displays the SDDI Rate. Within these figures, a lower ADDI rate is indicative of safer medication combinations, whereas a higher SDDI rate suggests greater effectiveness of the medication combinations.

The comparative baseline methods can be broadly divided into three distinct groups:

Instance-based methods: Methods such as Logistic Regression (LR), Ensemble of Classifier Chains (ECC) (Read et al., 2011), and LEAP (Zhang et al., 2017) focus primarily on the patient's current health condition. While this approach is straightforward, it often results in suboptimal predictive accuracy due to its reliance on present visit records alone, overlooking the valuable historical information in the patient's longitudinal history health records.

Longitudinal representation methods: In contrast, methods that emphasize longitudinal representation patterns, such as RETAIN (Choi, Bahadori, Sun et al., 2016) and GAMENet (Shang, Xiao et al., 2019), typically demonstrate superior performance. Compared to instance-based methods, these methods incorporate comprehensive longitudinal health data, underlining the importance of a holistic longitudinal approach in medication recommendation models. However, RETAIN overlooks drug–drug interactions (DDIs) relations, potentially leading to higher DDI rates and recommending riskier medication combinations. Similarly, GAMENet, heavily reliant on historical combinations, may yield a high DDI rate if the ground-truth history involves such combinations. Therefore, despite these methods showing enhanced performance, addressing adverse DDIs is imperative to ensure the safety of medication recommendations. Therefore, despite these methods showing enhanced performance, addressing adverse DDIs is imperative to ensure the safety of medication recommendations.

Knowledge integration methods: Approaches like SafeDrug (Yang, Xiao, Ma et al., 2021), MoleRec (Yang et al., 2023), and 4SDrug (Tan et al., 2022) integrate external knowledge to address the DDI issues and significantly improve accuracy and representation ability. SafeDrug utilizes molecular structures in medication encoding, balancing recommendation accuracy and low ADDI rates. This is attributed to its strategic sacrifice of recommendation accuracy in exchange for low ADDI values,

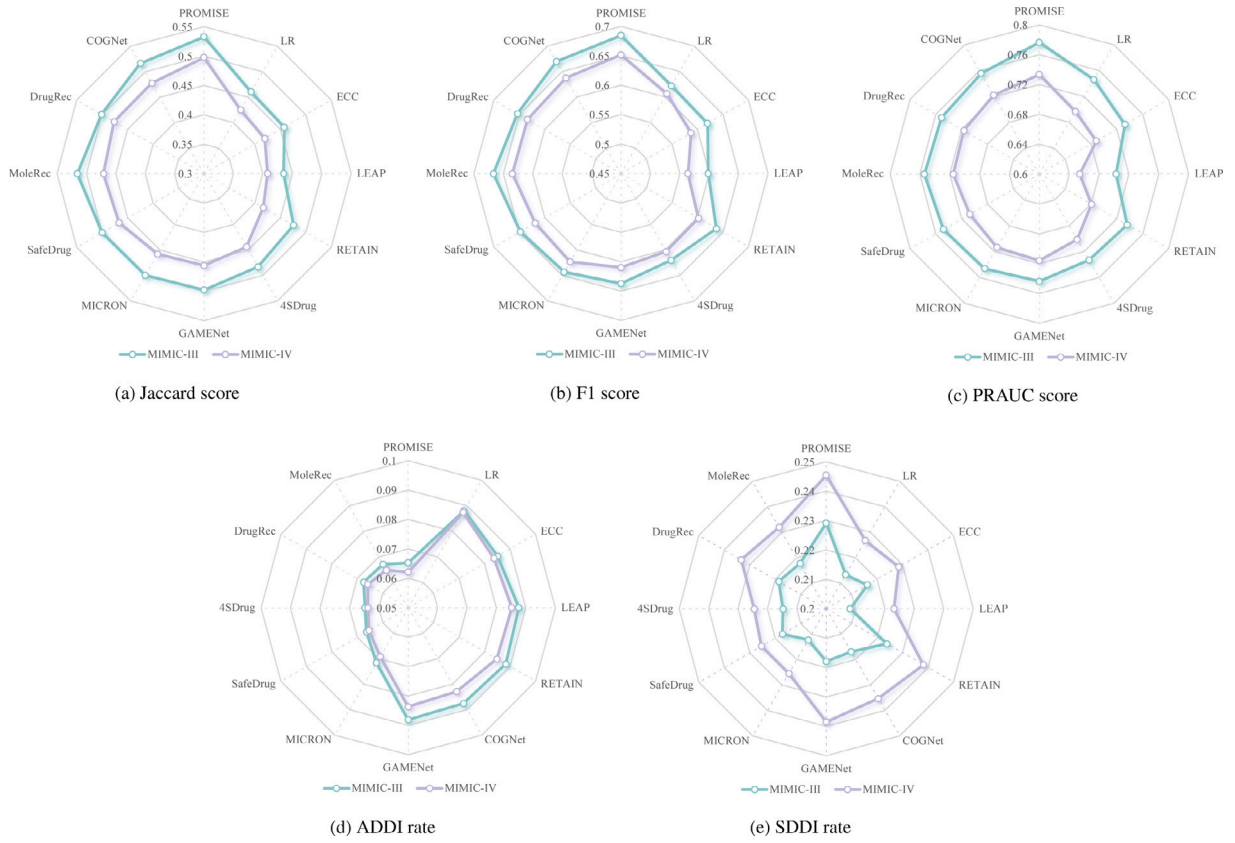


Fig. 8. The results of comparative experiments on the MIMIC-III and MIMIC-IV datasets. Our proposed PROMISE framework is compared with several baseline models on (a) Jaccard, (b) F1, (c) PRAUC, (d) ADDI Rate, and (e) SDDI Rate, evaluation metrics.

optimizing for accuracy only when below the target DDI. MoleRec advances further by modeling the dependencies between patient health conditions and molecular substructures. 4SDrug, employing a set-to-set comparison module for symptom-based medication combination prediction. While excelling in recommending small medication sets and reducing recommendation sizes, 4SDrug sacrifices accuracy compared to other models. Additionally, MICRON (Yang et al., 2021) and COGNet (Wu, Qiu et al., 2022) concentrate on analyzing changes of historical medication, with COGNet implementing a 'copy-or-predict' mechanism. While this improves performance, a high DDI rate remains due to the lack of DDI constraints. DrugRec (Sun et al., 2022) introduces a causal graph to reduce recommendation bias and integrates clinical symptoms for more accurate recommendations. Overall, these models highlight the crucial role of external knowledge, a principle also employed in our PROMISE, contributing to enhanced representativeness and accuracy in medication recommendations.

As indicated in Table 3, our proposed framework, PROMISE, demonstrates superior performance compared to other state-of-the-art methods, achieving the highest scores in the Jaccard, F1, and PRAUC metrics. **(1) Performance on the MIMIC-III Dataset**: PROMISE shows improvements of 1.61% in Jaccard, 1.64% in F1, and 2.06% in PRAUC score, compared to the leading method. It achieves a comparable ADDI rate to 4SDrug, which records the lowest ADDI rate, and shows a 1.05% enhancement in the SDDI rate. **(2) Performance on the MIMIC-IV Dataset**: Compared to the best-performing method on this dataset, PROMISE records improvements of 1.97% in Jaccard, 1.42% in F1, and 1.14% in PRAUC score. Our framework also exhibits a 0.16% reduction in the ADDI rate and a 0.69% improvement in the SDDI rate. These results highlight the effectiveness of PROMISE in delivering accurate, safe, and effective medication recommendations, outperforming existing methods in key performance metrics across both datasets.

Additionally, the statistical significance tests conducted across the MIMIC-III and MIMIC-IV datasets further validate the superior performance of PROMISE compared to state-of-the-art methods. **(1) MIMIC-III Dataset statistical significance**: The p-values for the Jaccard, F1, PRAUC, ADDI rate, and SDDI rate are 3e−9, 2e−11, 3e−7, 2e−10, and 7e−12, respectively. **(2) MIMIC-IV Dataset statistical significance**: The corresponding p-values are 2e−8, 5e−12, 9e−7, 5e−9, and 3e−11. These p-values, all significantly less than the conventional threshold of 0.05, confirm the statistical significance of PROMISE's superiority in performance metrics. This comprehensive statistical analysis conclusively establishes PROMISE's effectiveness in medication recommendation, highlighting its statistically significant improvements over existing methods.

Table 3
The results of comparative experiments on the MIMIC-III and MIMIC-IV datasets. The **boldface** is used to highlight the highest scores achieved, the underline is used to mark the best results among the compared methods.

	Method	ADDI ↓	SDDI ↑	Jaccard ↑	F1 ↑	PRAUC ↑	Avg. Med.
MIMIC-III	LR	$0.0881_{0.0008}$	$0.2133_{0.0015}$	$0.4604_{0.0026}$	$0.6218_{0.0026}$	$0.7461_{0.0024}$	$18.7424_{0.0738}$
	ECC	$0.0851_{0.0011}$	$0.2162_{0.0013}$	$0.4576_{0.0021}$	$0.6198_{0.0024}$	$0.7326_{0.0026}$	$17.1971_{0.1005}$
	LEAP	$0.0876_{0.0007}$	$0.2082_{0.0010}$	$0.4353_{0.0021}$	$0.5982_{0.0021}$	$0.7028_{0.0025}$	$18.8053_{0.0721}$
	RETAIN	$0.0883_{0.0015}$	$0.2238_{0.0026}$	$0.4757_{0.0027}$	$0.6373_{0.0027}$	$0.7359_{0.0038}$	$19.5626_{0.0810}$
	GAMENet	$0.0881_{0.0005}$	$0.2179_{0.0017}$	$0.4978_{0.0028}$	$0.6372_{0.0025}$	$0.7438_{0.0021}$	$26.7881_{0.2208}$
	4SDrug	$\mathbf{0.0647}_{0.0004}$	$0.2117_{0.0012}$	$0.4932_{0.0016}$	$0.6304_{0.0017}$	$0.7381_{0.0015}$	$15.3475_{0.0742}$
	SafeDrug	$0.0664_{0.0004}$	$0.2172_{0.0013}$	$0.5003_{0.0017}$	$0.6478_{0.0017}$	$0.7483_{0.0026}$	$20.7939_{0.0628}$
	MoleRec	$0.0671_{0.0006}$	$0.2178_{0.0014}$	$0.5156_{0.0021}$	$0.6669_{0.0020}$	$0.7541_{0.0019}$	$21.1318_{0.1213}$
	MICRON	$0.0716_{0.0006}$	$0.2122_{0.0028}$	$0.4996_{0.0027}$	$0.6435_{0.0024}$	$0.7463_{0.0022}$	$17.7604_{0.0989}$
	COGNet	$0.0876_{0.0008}$	$0.2169_{0.0013}$	$\underline{0.5165}_{0.0023}$	$\underline{0.6684}_{0.0025}$	$\underline{0.7558}_{0.0030}$	$28.3909_{0.2008}$
	DrugRec	$0.0675_{0.0006}$	$\underline{0.2186}_{0.0015}$	$0.5012_{0.0015}$	$0.6533_{0.0021}$	$0.7512_{0.0020}$	$20.5433_{0.0772}$
	PROMISE(Ours)	$\underline{0.0653}_{0.0006}$	$\mathbf{0.2291}_{0.0013}$	$\mathbf{0.5326}_{0.0018}$	$\mathbf{0.6848}_{0.0021}$	$\mathbf{0.7764}_{0.0020}$	$19.7645_{0.0921}$
	p-value	3e−9	2e−11	3e−7	2e−10	7e−12	/
MIMIC-IV	LR	$0.0874_{0.0007}$	$0.2267_{0.0012}$	$0.4248_{0.0023}$	$0.6063_{0.0021}$	$0.6956_{0.0019}$	$15.2629_{0.1002}$
	ECC	$0.0837_{0.0006}$	$0.2285_{0.0011}$	$0.4195_{0.0021}$	$0.5875_{0.0019}$	$0.6883_{0.0022}$	$14.1902_{0.0975}$
	LEAP	$0.0852_{0.0007}$	$0.2231_{0.0012}$	$0.4081_{0.0021}$	$0.5638_{0.0017}$	$0.6541_{0.0023}$	$18.1347_{0.0857}$
	RETAIN	$0.0848_{0.0006}$	$0.2382_{0.0008}$	$0.4161_{0.0016}$	$0.6022_{0.0017}$	$0.6806_{0.0018}$	$18.1109_{0.0713}$
	GAMENet	$0.0837_{0.0004}$	$\underline{0.2385}_{0.0006}$	$0.4563_{0.0017}$	$0.6098_{0.0016}$	$0.7159_{0.0016}$	$18.0648_{0.0781}$
	4SDrug	$\underline{0.0637}_{0.0004}$	$0.2195_{0.0012}$	$0.4539_{0.0011}$	$0.6034_{0.0010}$	$0.7011_{0.0011}$	$12.5213_{0.0665}$
	SafeDrug	$0.0654_{0.0004}$	$0.2234_{0.0014}$	$0.4669_{0.0011}$	$0.6186_{0.0012}$	$0.7074_{0.0009}$	$13.5801_{0.0627}$
	MoleRec	$0.0648_{0.0006}$	$0.2320_{0.0011}$	$0.4711_{0.0012}$	$0.6348_{0.0015}$	$0.7153_{0.0011}$	$17.0012_{0.0898}$
	MICRON	$0.0691_{0.0008}$	$0.2264_{0.0013}$	$0.4579_{0.0021}$	$0.6233_{0.0023}$	$0.7134_{0.0018}$	$16.7695_{0.0754}$
	COGNet	$0.0829_{0.0011}$	$0.2353_{0.0010}$	$\underline{0.4776}_{0.0020}$	$\underline{0.6375}_{0.0016}$	$\underline{0.7221}_{0.0019}$	$19.3678_{0.0836}$
	DrugRec	$0.0659_{0.0006}$	$0.2334_{0.0009}$	$0.4765_{0.0008}$	$0.6338_{0.0011}$	$0.7165_{0.0010}$	$16.4512_{0.0651}$
	PROMISE(Ours)	$\mathbf{0.0621}_{0.0007}$	$\mathbf{0.2454}_{0.0009}$	$\mathbf{0.4973}_{0.0010}$	$\mathbf{0.6517}_{0.0008}$	$\mathbf{0.7335}_{0.0010}$	$17.1309_{0.0741}$
	p-value	2e−8	5e−12	9e−7	5e−9	3e−11	/

Table 4
Ablation study results on the MIMIC-III and MIMIC-IV dataset.

	Method	ADDI ↓	SDDI ↑	Jaccard ↑	F1 ↑	PRAUC ↑	Avg. Med
MIMIC-III	PROMISE w/o CodeInit	$0.0713_{0.0007}$	$0.2165_{0.0016}$	$0.5261_{0.0015}$	$0.6734_{0.0021}$	$0.7698_{0.0018}$	$20.1964_{0.1054}$
	PROMISE w/o OntInit	$0.0688_{0.0006}$	$0.2268_{0.0014}$	$0.5298_{0.0011}$	$0.6796_{0.0017}$	$0.7711_{0.0019}$	$20.3056_{0.0768}$
	PROMISE w/o MoleInit	$0.0707_{0.0007}$	$0.2236_{0.0014}$	$0.5283_{0.0012}$	$0.6769_{0.0020}$	$0.7682_{0.0013}$	$19.6790_{0.1198}$
	PROMISE w/o Pre	$0.0687_{0.0006}$	$0.2216_{0.0017}$	$0.4902_{0.0018}$	$0.6427_{0.0023}$	$0.7389_{0.0020}$	$20.3017_{0.0932}$
	PROMISE w/o $\mathcal{L}_{\text{MP}}$	$0.0713_{0.0007}$	$0.2114_{0.0016}$	$0.5140_{0.0023}$	$0.6607_{0.0025}$	$0.7434_{0.0022}$	$19.6703_{0.0781}$
	PROMISE w/o $\mathcal{L}_{\text{MI}}$	$0.0667_{0.0009}$	$0.2235_{0.0015}$	$0.5252_{0.0024}$	$0.6784_{0.0027}$	$0.7680_{0.0024}$	$20.3951_{0.1223}$
	PROMISE w/o Aug	$0.0701_{0.0008}$	$0.2156_{0.0014}$	$0.5196_{0.0016}$	$0.6654_{0.0023}$	$0.7467_{0.0017}$	$19.5247_{0.1301}$
	PROMISE-C	$0.0676_{0.0010}$	$0.2236_{0.0015}$	$0.5236_{0.0018}$	$0.6779_{0.0021}$	$0.7715_{0.0021}$	$19.8017_{0.0956}$
	PROMISE-T	$0.0741_{0.0008}$	$0.2056_{0.0014}$	$0.5065_{0.0017}$	$0.6521_{0.0019}$	$0.7553_{0.0018}$	$19.8132_{0.0981}$
	PROMISE	$0.0653_{0.0006}$	$0.2291_{0.0013}$	$0.5326_{0.0018}$	$0.6848_{0.0021}$	$0.7764_{0.0020}$	$19.7645_{0.0921}$
MIMIC-IV	PROMISE w/o CodeInit	$0.0724_{0.0007}$	$0.2103_{0.0017}$	$0.4913_{0.0023}$	$0.6406_{0.0018}$	$0.7251_{0.0017}$	$14.7263_{0.1124}$
	PROMISE w/o OntInit	$0.0681_{0.0008}$	$0.2195_{0.0021}$	$0.4966_{0.0020}$	$0.6465_{0.0017}$	$0.7263_{0.0016}$	$13.8957_{0.0962}$
	PROMISE w/o MoleInit	$0.0716_{0.0007}$	$0.2169_{0.0016}$	$0.4943_{0.0021}$	$0.6439_{0.0016}$	$0.7236_{0.0016}$	$14.3128_{0.0827}$
	PROMISE w/o Pre	$0.0680_{0.0006}$	$0.2149_{0.0017}$	$0.4587_{0.0023}$	$0.6121_{0.0018}$	$0.6978_{0.0018}$	$13.5369_{0.1275}$
	PROMISE w/o $\mathcal{L}_{\text{MP}}$	$0.0706_{0.0007}$	$0.2050_{0.0015}$	$0.4800_{0.0025}$	$0.6285_{0.0021}$	$0.7002_{0.0020}$	$13.6174_{0.1201}$
	PROMISE w/o $\mathcal{L}_{\text{MI}}$	$0.0647_{0.0008}$	$0.2186_{0.0019}$	$0.4908_{0.0026}$	$0.6438_{0.0018}$	$0.7275_{0.0022}$	$13.7649_{0.0854}$
	PROMISE w/o Aug	$0.0694_{0.0008}$	$0.2091_{0.0016}$	$0.4852_{0.0017}$	$0.6330_{0.0016}$	$0.7033_{0.0015}$	$14.0093_{0.0806}$
	PROMISE-C	$0.0651_{0.0007}$	$0.2169_{0.0015}$	$0.4890_{0.0019}$	$0.6449_{0.0015}$	$0.7267_{0.0018}$	$14.8147_{0.0729}$
	PROMISE-T	$0.0734_{0.0009}$	$0.1994_{0.0013}$	$0.4730_{0.0020}$	$0.6203_{0.0017}$	$0.7114_{0.0016}$	$13.8512_{0.1224}$
	PROMISE	$0.0621_{0.0007}$	$0.2254_{0.0015}$	$0.4973_{0.0019}$	$0.6517_{0.0016}$	$0.7335_{0.0015}$	$14.1309_{0.1541}$

5.2. Ablation studies results

To better understand the contributions of each component in our proposed PROMISE framework, we carried out ablation studies. These studies involved creating and evaluating different variants of PROMISE, each omitting one or more of its modules. The variants used in these ablation studies included:

- PROMISE w/o CodeInit means without the code initialization module.
- PROMISE w/o OntInit means without the diagnosis and procedure code initialization step.
- PROMISE w/o MoleInit means without the medication code initialization step.
- PROMISE w/o Pre means we train PROMISE from scratch based on the fine-tuning setting.
- PROMISE w/o $\mathcal{L}_{\text{VP}}$ means without the modality-specific prediction loss function.
- PROMISE w/o $\mathcal{L}_{\text{MI}}$ means without the modality-augment learning loss function.
- PROMISE w/o Aug means without the multimodal augmentation module.
- PROMISE-C means the text-level information is not used for model fine-tuning.
- PROMISE-T means the code-level information is not used for model fine-tuning.

Table 4 presents the results of ablation studies conducted in three sets of experiments on different datasets.

Effect of code initialization module: In the first set of our ablation studies, we focused on evaluating the impact of the Code Initialization Module and its individual components on the overall effectiveness of our framework. The results on both datasets in Table 4 indicate that omitting the pretraining of the initial diagnosis and procedure codes module (PROMISE w/o OntInit) or the initial medication representations module (PROMISE w/o MoleInit) adversely affects the model's performance. In these cases, the sub-hypergraph modules encoding structured patient representations rely on randomly initialized node embeddings, negatively impacting model performance. Without the medication representation constrained by substructure and DDI knowledge, the DDI rate tends to increase unfavorably. Further, our framework constrains both two DDI rates and prediction accuracy in the loss function, causing the framework sacrifices performance to ensure an appropriate DDI value.

Role of different loss functions in pre-training: In the second set of experiments, our objective was to analyze how various loss functions used during the pre-training stage affect the performance of PROMISE. It is evident from the result that both $\mathcal{L}_{\text{MP}}$ and $\mathcal{L}_{\text{MI}}$ play a crucial role in constraining the representation of multimodal patient representations in the shared feature space and the mutual information between multimodal patient representations, thereby assisting the model in obtaining a comprehensive patient representation. From this set of experiments, we can observe that, firstly, the absence of the loss function that maximizes the mutual information of multimodal representation makes it challenging for our framework to eliminate redundant information and align representations in different modalities. Secondly, removing the constraint on the distance between the representation of different modal mappings and the target in the shared feature space also adversely affects the model's performance. Lastly, if the entire pre-training part is omitted, and the framework is trained with fine-tuning loss in an end-to-end method, the model's performance decreases. Compared with some end-to-end models, the results of PROMISE demonstrate the effectiveness of its multimodal framework.

Impact of different modalities on performance: In the third set of our experiments, we focused on evaluating the influence of different modalities on the performance of the PROMISE framework. From the result, we observe that PROMISE-C outperforms PROMISE-T on all datasets. This suggests that structural code information of medical events is more predictive of clinical outcomes than semantic medical text. The addition of medical text modality significantly improved performance. However, without fusing both modalities (PROMISE w/o Aug), the results were suboptimal, underscoring the importance of multimodal representation fusion.

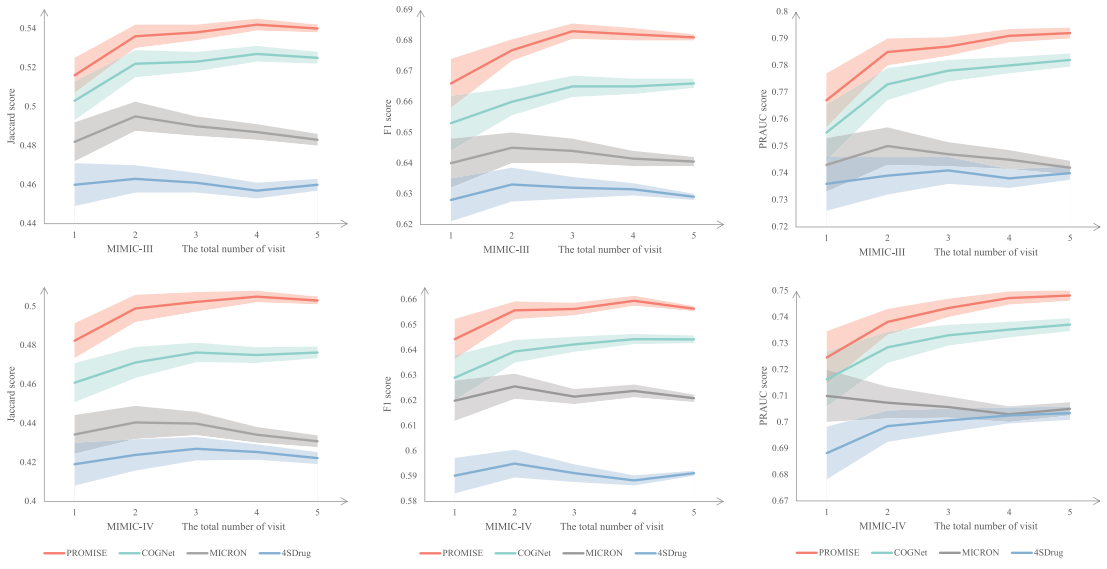
In summary, each component of PROMISE plays a significant role in enhancing precision. Excluding any modality disrupts the learning process of patient representation, affecting the model's ability to discern valid correlations. This emphasizes the efficacy of leveraging multimodal information in medical codes for optimized medication recommendation.

5.3. Influences of the number of visits

To delve deeper into PROMISE's ability to effectively use historical medical data, we conducted an analysis that focused on the impact of varying numbers of total patient visits on PROMISE's performance across both MIMIC-III and MIMIC-IV datasets. Recognizing that the average number of visits per patient is below three, as shown in Table 2, and noting that less than 10% of patients have more than five visits, we stratified our datasets based on the total number of patient visits. This stratification allowed for a more detailed and nuanced assessment of how visit frequency influences the model's performance. The results of this comparison, based on different visit frequencies, are illustrated in Fig. 9.(a–c). For our evaluation, we selected Jaccard, F1 score, and PRAUC as the evaluation metrics. The first row of this figure presents the results from the MIMIC-III dataset, while the second row shows the results from MIMIC-IV. In each subgraph, the horizontal axis represents the length of the patient's visit history (number of visits), and the vertical axis displays various performance metrics.

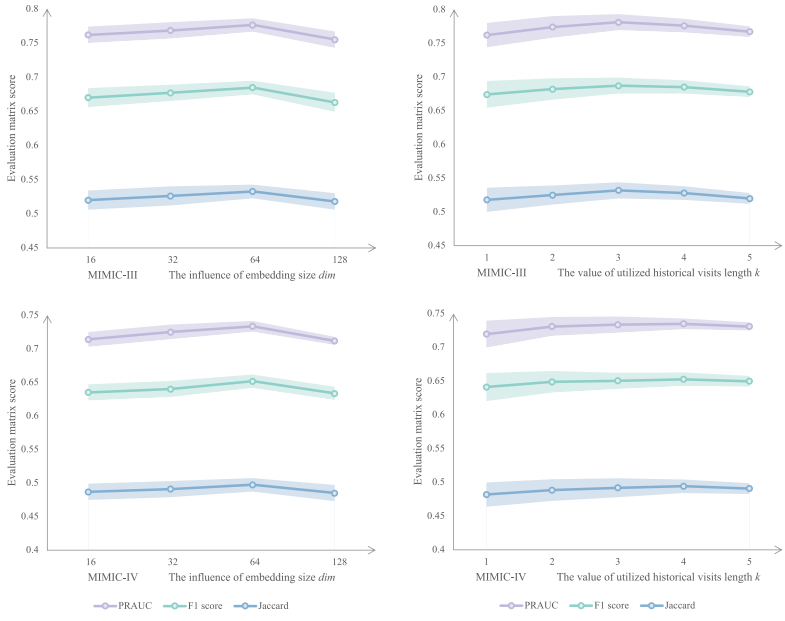
We chose COGNet (Wu, Qiu et al., 2022), MICRON (Yang et al., 2021), and 4SDrug (Tan et al., 2022) as baseline comparisons. Among these, COGNet effectively utilizes historical medication information and achieves the best performance of existing methods. This visualization offers a clear depiction of how PROMISE's effectiveness varies with the extent of a patient's visit history. The results demonstrate PROMISE's consistent superiority across various visit times, demonstrating superior performance as the number of visits increases. This trend underscores PROMISE's proficiency in harnessing multimodal information for precise recommendations, especially for shorter visit lengths. In contrast, while 4SDrug's performance remained stable as its small set of recommendations is more of an instance-based approach, MICRON showed a declining trend due to error accumulation from its iterative updates.

These findings suggest that while incorporating historical records is beneficial, overly extensive historical data can adversely affect the model's current predictions. This occurs because PROMISE adopts the same prediction method based on multiple visits as previous work, which ignores the time interval between different visits.



(a) Jaccard score (b) F1 score (c) PRAUC score

Fig. 9. The performance of different number of visits. The first row represents the results in the MIMIC-III dataset, and the second row represents the results in the MIMIC-IV dataset. Subfigure (a)–(c) show the Jaccard, F1, and PRAUC score for COGNet, MICRON, 4SDrug, and our PROMISE.

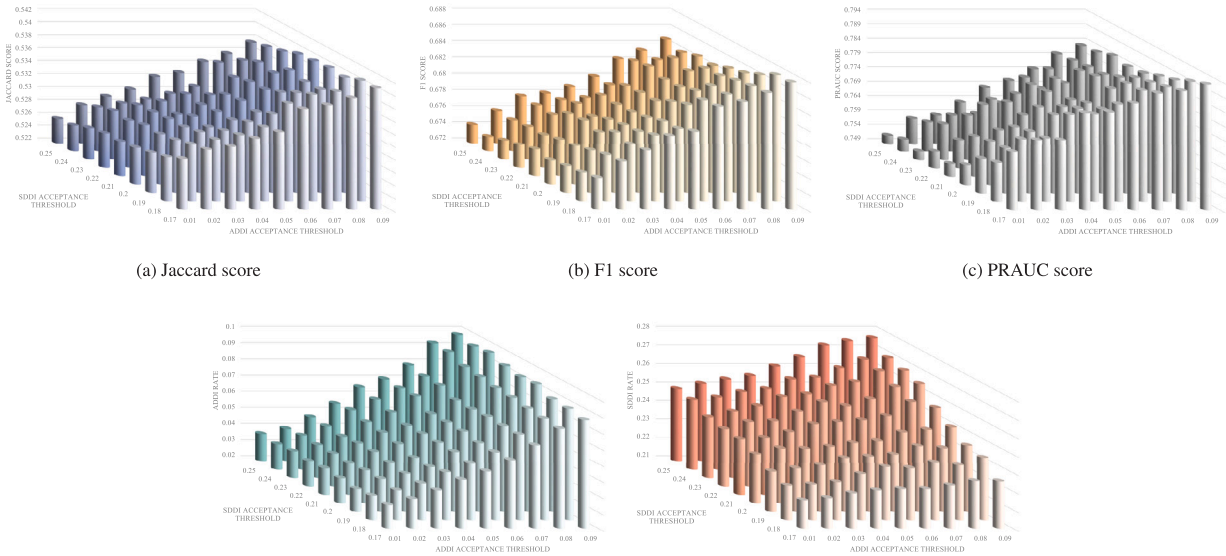


(a) The impact of *dim* (b) The impact of k

Fig. 10. Hyperparameter impact. Subfigure (a)–(b) shows the influence of different dim for PORMISE on two datasets, we considered values of dim within the set [16, 32, 64, 128], where dim represents for the embedding dimension of E_d, E_p, E_m. Subfigure (c)–(d) shows the influence of different k for PORMISE on two datasets, we considered values of k from 1 to 5, where k represents the utilized historical visit information.

5.4. Hyperparameter testing and analysis

In this subsection, we detail the optimization process for several critical hyperparameters within our framework, specifically for the embedding dimension (dim) of E_d, E_p, E_m within the set [16, 32, 64, 128], and the utilized historical visit value k within the range [1, 2, 3, 4, 5].



(a) Jaccard score (b) F1 score (c) PRAUC score

(d) ADDI rate (e) SDDI rate

Fig. 11. Sensitivity analysis of the *Controllable DDI* through varying thresholds combination: Exploring ADDI (γ_a) range 0.01–0.09 on the x-axis and SDDI (γ_s) range 0.17–0.25 on the y-axis. Evaluation of DDI Threshold Combinations Using Metrics (a) Jaccard, (b) F1, (c) PRAUC, (d) ADDI Rate, and (e) SDDI Rate on MIMIC-III Dataset.

Embedding Dimension (*dim*): As presented in Fig. 10.(a)–(b), we undertook a series of scaling experiments that involved adjustments to the embedding size to assess the influence of the embedding dimension (dim) on the performance of our proposed framework. The results demonstrate that the performance of PROMISE improves with an increase in the embedding dimension up to a certain point. A peak in performance was observed at a dimension of 64, beyond which further increases led to a marked decrease in model performance. This reduction in performance is attributed to the framework's increased noise susceptibility and the heightened risk of overfitting associated with excessive dimensionality. Based on these findings, we set the embedding dimension dim to 64.

Utilized historical visits length (k): As indicated in Fig. 9.(a–c), we observed that the highest prediction accuracy on both the MIMIC-III and MIMIC-IV datasets occurred for patients with three to four visits. This observation prompted further exploration into the impact of different numbers of the utilized historical visits length (k) on the performance of PROMISE. Our comparison experiments, which varied k on the two datasets, showed some intriguing results as shown in Fig. 10.(c)–(d): (1) In the MIMIC-III dataset, PROMISE's effectiveness increased as k rose from 1 to 3, but then slightly declined from $k = 4$ to 5. (2) In the MIMIC-IV dataset, the performance indicators of PROMISE peaked when k was set to 4.

Extending historical records beyond a certain length can introduce challenges such as irrelevant or misleading information, which may compromise the accuracy of predictive analyses. These challenges are compounded by increased computational demands and the necessity for additional data storage. Moreover, overly extended history lengths risk incorporating outdated or less relevant data into the model.

To mitigate these issues, it is advisable to select a moderate history length for inclusion in the analysis framework. As demonstrated in the results, setting the history length to $k = 3$ and $k = 4$ achieves optimal performance on two dataset, striking an effective balance between the benefits of comprehensive historical data and the need to maintain model efficiency and relevance. This strategy ensures accurate predictions by minimizing the inclusion of noise from extensive historical records, thereby achieving an optimal compromise between providing sufficient historical context and ensuring current applicability.

Through these optimizations, we aim to enhance the overall performance of our proposed model while addressing the computational and methodological challenges associated with hyperparameter selection.

5.5. Sensitivity analysis of DDIs thresholds combination

In evaluating the efficacy of our controllable DDI loss for regulating ADDI and SDDI rates in medication recommendations, we conducted experiments using a range of threshold combinations for ADDI (γ_a) and SDDI (γ_s) in the training of PROMISE. The ADDI thresholds were set between 0.01 and 0.09, while SDDI thresholds ranged from 0.17 to 0.25. Considering the inherent ADDI and SDDI rates in the MIMIC III dataset, which are 0.813 and 0.1583 respectively in Table 2, our objective was to reduce the ADDI rates and increase the SDDI rates in our recommendations.

Visit	Method	ADDI	SDDI	Correct	Missed	Unseen
1st	PROMISE	0.0430	0.3053	21	5	6: R05C$_{(S\#3)}$, A03F$_{(S\#7)}$, A04A, N05A$_{(S\#8)}$ J01X$_{(A\#1,S\#7)}$, D07A$_{(A\#6,S\#9)}$, N06A$_{(A\#6)}$
	COGNet	0.0766	0.2137	20	6	13: J01M$_{(S\#2)}$, R05C$_{(S\#3)}$, A12A, C07A$_{(A\#3,S\#2)}$, J01C$_{(A\#1)}$, A07E$_{(A\#4,S\#6)}$, N05B$_{(A\#2)}$, A04A, J01X$_{(A\#1,S\#7)}$, N03A$_{(A\#2,S\#6)}$, D07A$_{(A\#5,S\#8)}$, N06A$_{(A\#6)}$, N05A$_{(S\#7)}$
	MoleRec	0.0683	0.2044	19	7	10: J01M$_{(S\#2)}$, R05C$_{(S\#3)}$, D10A$_{(A\#4,S\#4)}$, J01F$_{(A\#1,S\#5)}$, A12A, J01C$_{(A\#1)}$, J01X$_{(A\#1,S\#7)}$, N03A$_{(A\#2,S\#6)}$, B02B$_{(S\#4)}$, N05A$_{(S\#4)}$
	4SDrug	0.0566	0.1823	19	7	8: J01M$_{(S\#2)}$, R05C$_{(S\#3)}$, C07A$_{(A\#3,S\#2)}$, N05B$_{(A\#2)}$, A04A, N03A$_{(A\#2,S\#6)}$, A12A, N06A$_{(A\#6)}$
2nd	PROMISE	0.0653	0.2278	15	2	3: J01D, C01B$_{(A\#1,S\#6)}$, C01C$_{(S\#2)}$
	COGNet	0.1075	0.1764	15	2	11: J01M$_{(S\#2)}$, N03A$_{(A\#1)}$, C07A$_{(A\#3,S\#2)}$, N02A$_{(A\#1,S\#6)}$, C01C, N05B$_{(A\#1)}$, A04A, A07E$_{(A\#3,S\#4)}$, N01A, N06A$_{(A\#6)}$, J01D
	MoleRec	0.0692	0.2215	14	3	4: J01D, A04A, M04A$_{(S\#1)}$, N02A$_{(A\#1,S\#6)}$
	4SDrug	0.0606	0.1638	14	3	8: J01M$_{(S\#2)}$, N03A$_{(A\#1)}$, N02A$_{(A\#1,S\#6)}$, C01B$_{(S\#5)}$, N05B$_{(A\#1)}$, A04A, A07E$_{(A\#3,S\#4)}$, J01D
3rd	PROMISE	0.0598	0.2367	23	2	10: M04A$_{(A\#2,S\#1)}$, J01M$_{(S\#6)}$, R05C$_{(S\#4)}$, N05A$_{(S\#10)}$, A02A$_{(S\#3)}$, N02A$_{(A\#1,S\#12)}$, C01B$_{(A\#1,S\#8)}$, N01A$_{(S\#10)}$, D07A$_{(A\#9,S\#10)}$, A12A
	COGNet	0.0966	0.2292	20	5	13: J01M$_{(S\#5)}$, R05C$_{(S\#3)}$, N03A$_{(A\#1,S\#10)}$, A12A, A02A$_{(S\#3)}$, N02A$_{(A\#1,S\#10)}$, C01B$_{(A\#1,S\#7)}$, N05B$_{(A\#1)}$, A07E$_{(A\#5,S\#5)}$, N01A$_{(S\#8)}$, D07A$_{(A\#8,S\#9)}$, N06A$_{(A\#8)}$, N05A$_{(S\#8)}$
	MoleRec	0.0719	0.2365	20	5	7: B02B$_{(S\#4)}$, C03A$_{(A\#1)}$, C01B$_{(A\#1,S\#7)}$, N02A$_{(A\#1,S\#10)}$, C03B$_{(S\#6)}$, N01A$_{(S\#8)}$, A12A
	4SDrug	0.0584	0.1907	19	6	7: J01M$_{(S\#4)}$, R05C$_{(S\#3)}$, C01B$_{(S\#6)}$, N02A$_{(A\#1,S\#9)}$, N01A$_{(S\#7)}$, A12A, N06A$_{(A\#7)}$
4th	PROMISE	0.0543	0.1920	18	3	6: N05C$_{(A\#2,S\#5)}$, J01M$_{(S\#6)}$, C07A$_{(A\#2,S\#2)}$, A04A, J01D, N05A$_{(S\#7)}$
	COGNet	0.1008	0.2137	18	3	14: N05C$_{(A\#2,S\#5)}$, J01M$_{(S\#6)}$, R05C$_{(S\#3)}$, N03A$_{(A\#2,S\#9)}$, A12A, J01D, C07A$_{(A\#3,S\#2)}$, C01B$_{(A\#1,S\#7)}$, A04A, A07E$_{(A\#5,S\#5)}$, N01A$_{(S\#7)}$, D07A$_{(A\#6,S\#8)}$, N06A$_{(A\#7)}$, N05A$_{(S\#7)}$
	MoleRec	0.0652	0.1377	15	6	9: M04A$_{(S\#1)}$, J01M$_{(S\#4)}$, R05C$_{(S\#3)}$, C07A$_{(A\#2,S\#2)}$, S01E$_{(A\#3,S\#4)}$, A03F$_{(S\#5)}$, A04A, J01D, A12A
	4SDrug	0.0700	0.1833	15	6	10: N05C$_{(A\#2,S\#5)}$, J01M$_{(S\#5)}$, R05C$_{(S\#3)}$, N03A$_{(A\#2,S\#8)}$, A12A, C01B$_{(S\#6)}$, A04A, J01D, C07A$_{(A\#2,S\#2)}$, N06A$_{(A\#5)}$
5th	PROMISE	0.0741	0.2792	19	4	8: M03A, J01D$_{(A\#1)}$, N02A$_{(A\#1,S\#11)}$, C01B$_{(A\#1,S\#7)}$, C02D$_{(S\#10)}$, S01E$_{(A\#5,S\#8)}$, N01A$_{(S\#9)}$, A12A
	COGNet	0.0819	0.2193	18	5	15: A03F$_{(A\#1,S\#8)}$, J01M$_{(S\#5)}$, R05C$_{(S\#3)}$, N03A$_{(A\#2,S\#11)}$, A12A, M03A, J01D$_{(A\#1)}$, N02A$_{(A\#1,S\#11)}$, C01B$_{(A\#1,S\#8)}$, D01A, N05B$_{(A\#1)}$, A04A, N01A$_{(S\#8)}$, D07A$_{(A\#8,S\#10)}$, N06A$_{(A\#8)}$
	MoleRec	0.0504	0.2258	17	6	15: M04A$_{(A\#1,S\#1)}$, J01M$_{(S\#4)}$, N01A$_{(S\#7)}$, A12A, J01D, M03A, D06A$_{(S\#5)}$, N02A$_{(A\#1,S\#9)}$, S01E$_{(A\#4,S\#6)}$, A04A A03F$_{(S\#6)}$, J01X$_{(S\#6)}$, C01B$_{(S\#6)}$, R05C$_{(S\#3)}$, D07A$_{(A\#6,S\#8)}$
	4SDrug	0.0733	0.2400	17	6	8: J01M$_{(S\#4)}$, N01A$_{(S\#7)}$, N03A$_{(A\#1,S\#9)}$, C01B$_{(S\#6)}$, N02A$_{(A\#1,S\#9)}$, J01D$_{(A\#1)}$, A12A, N06A$_{(A\#7)}$

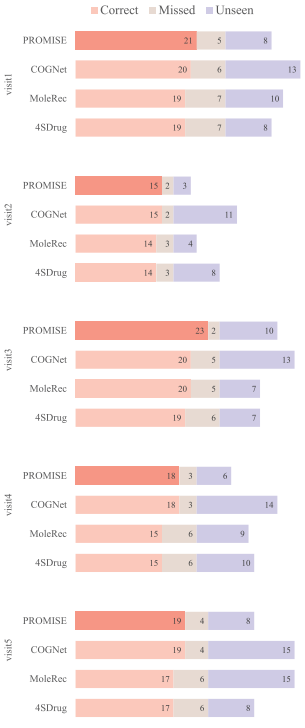


Fig. 12. Example of medication recommendations for patient A in MIMIC-III with multiple visits.

To mitigate the risk of over-prescribing, we established an upper limit for the SDDI rate at 0.25. Subsequently, we assessed the impact of each threshold pairing, monitoring shifts in various metrics within a coordinate framework. The mean values for these metrics are presented in Fig. 11.

Each sub-figure in Fig. 11 illustrates the variation in specific indicators with changes in the ADDI (γ_a) and SDDI (γ_s) thresholds. For example:

- Increasing γ_s while keeping γ_a constant worsens the ADDI rate (Fig. 11.(d)) and decreases overall recommendation performance (Fig. 11.(a–c)). This suggests that a too-high SDDI threshold can lead to unnecessary medication recommendations.
- Conversely, lowering γ_a while maintaining γ_s leads to a significant drop in both the ADDI rate and other performance metrics (Fig. 11.(a)–(c)). This is due to the inherent ADDIs in the dataset; reducing γ_a restricts certain medications from being recommended, causing deviation from the actual data.

These findings corroborate the effectiveness of our controllable DDI loss in precisely regulating ADDI and SDDI rates within medication recommendations. Employing this loss function allows us to guarantee that our recommendations not only exhibit a reduced ADDI rate relative to the baseline but also maintain the SDDI rate within a predetermined boundary.

5.6. Case studies

We present case studies with analyzing two patients from the MIMIC-III test set to demonstrate the effectiveness of PROMISE. Each patient in our case studies including the clinical record of five visits.

For comparison, we also employed COGNet (Wu, Qiu et al., 2022), MoleRec (Yang et al., 2023), and 4SDrug (Tan et al., 2022) to predict medication combinations. The outcomes of these predictions are illustrated in Figs. 12 and 13, providing a clear and visual summary of the results.

In both figures, the "Correct" column represents medication sets that align with the ground truth medication set. The "Missed" column highlights medications that are omitted in the model predictions. The "Unseen" column lists the set of medication combination predicted by the models but do not exist in the ground truth. Each case study is visually divided into two sections: the

Visit	Method	ADDI	SDDI	Correct	Missed	Unseen
1st	PROMISE	0.0538	0.2854	23	5	3: $N06A_{(A\#6)}$, $J01C_{(S\#12)}$, $A01A_{(A\#2,S\#15)}$
	COGNet	0.0817	0.2151	24	4	10: $A01A_{(A\#3,S\#12)}$, $J01X_{(A\#1,S\#7)}$, $N06A_{(A\#6)}$, $C01C_{(S\#2)}$, $N05B_{(A\#2)}$, $H03A$, $A07E_{(A\#5,S\#3)}$, $N05A_{(A\#1,S\#8)}$, $R01A_{(A\#4,S\#9)}$, $J01C_{(A\#1,S\#8)}$
	MoleRec	0.0677	0.1760	21	7	7: $J01X_{(A\#1,S\#7)}$, $N05A_{(A\#1,S\#11)}$, $N06A_{(A\#6)}$, $C03A_{(A\#2)}$, $A01A_{(A\#2,S\#14)}$, $B02B_{(S\#11)}$, $N05B_{(A\#2)}$
	4SDrug	0.0716	0.2000	14	14	1: $A01A_{(A\#2,S\#7)}$
2nd	PROMISE	0.0724	0.3115	18	2	5: $C02D_{(S\#9)}$, $A12B_{(A\#1)}$, $J01M_{(S\#7)}$, $C03B_{(S\#9)}$, $A01A_{(A\#3,S\#11)}$
	COGNet	0.0869	0.2336	17	3	9: $C02D_{(S\#8)}$, $A12B_{(A\#1)}$, $J01M_{(S\#6)}$, $H03A$, $N05A_{(A\#1,S\#7)}$, $N06A_{(A\#7)}$, $A02A_{(S\#2)}$, $C01B_{(A\#1,S\#7)}$, $A01A_{(A\#3,S\#10)}$
	MoleRec	0.0769	0.2318	17	3	6: $C02D_{(S\#8)}$, $A12B_{(A\#1)}$, $A11C_{(A\#1)}$, $M04A_{(A\#1,S\#2)}$, $C01B_{(A\#1,S\#7)}$, $A01A_{(A\#3,S\#10)}$
	4SDrug	0.0762	0.2157	12	8	3: $A01A_{(A\#2,S\#7)}$, $A12B_{(A\#1)}$, $J01M_{(S\#5)}$
3rd	PROMISE	0.0723	0.2738	20	6	6: $A12B_{(A\#1)}$, $R03A_{(A\#1)}$, $N03A_{(A\#3,S\#11)}$, $J01D_{(A\#1)}$, $A01A_{(A\#1,S\#13)}$, $C09A_{(A\#6,S\#13)}$
	COGNet	0.0831	0.2569	16	10	7: $A12B_{(A\#1)}$, $N03A_{(A\#1,S\#9)}$, $H03A$, $R03A_{(A\#1)}$, $J01D_{(A\#1)}$, $A01A_{(A\#1,S\#10)}$, $C09A_{(A\#6,S\#10)}$
	MoleRec	0.0788	0.2517	16	10	7: $A12B_{(A\#2)}$, $N03A_{(S\#10)}$, $A10B_{(A\#2,S\#11)}$, $R03A_{(A\#1)}$, $J01D_{(A\#1)}$, $A01A_{(A\#1,S\#11)}$, $C09A_{(A\#6,S\#11)}$
	4SDrug	0.0762	0.2571	12	14	3: $A01A_{(A\#1,S\#8)}$, $A12B_{(A\#1)}$, $R03A$
4th	PROMISE	0.0804	0.3095	16	3	5: $A12B_{(A\#1)}$, $A03B_{(S\#10)}$, $R03A_{(A\#1)}$, $A07A_{(S\#10)}$, $C09A_{(A\#7,S\#10)}$
	COGNet	0.0865	0.2619	14	5	14: $A12B_{(A\#1)}$, $J01M_{(S\#6)}$, $N05A_{(S\#6)}$, $N05B_{(A\#2)}$, $A03B_{(S\#9)}$, $N06A_{(A\#5)}$, $R03A_{(A\#1)}$, $A04A$, $A02A_{(S\#3)}$, $A07A_{(S\#8)}$, $J01D_{(A\#1)}$, $C01B_{(A\#1,S\#5)}$, $R01A_{(A\#3,S\#8)}$, $C09A_{(A\#6,S\#9)}$
	MoleRec	0.0910	0.2557	15	4	8: $A12B_{(A\#1)}$, $A03B_{(S\#10)}$, $A10B_{(A\#2,S\#10)}$, $C02A_{(A\#1,S\#10)}$, $M04A_{(A\#1,S\#3)}$, $A04A$, $A02A_{(S\#3)}$, $C09A_{(A\#6,S\#10)}$
	4SDrug	0.0897	0.2051	10	9	3: $A12B_{(A\#1)}$, $A03B_{(S\#6)}$, $R03A_{(A\#1)}$
5th	PROMISE	0.0556	0.2792	22	4	6: $N05C_{(A\#2,S\#5)}$, $A12B_{(A\#1)}$, $N03A_{(A\#2,S\#10)}$, $R03A_{(A\#1)}$, $R01A_{(A\#3,S\#10)}$, $A07A_{(S\#10)}$
	COGNet	0.0851	0.2345	21	5	8: $N05C_{(A\#2,S\#5)}$, $A12B_{(A\#1)}$, $V03A_{(A\#1,S\#2)}$, $N06A_{(A\#5)}$, $R03A_{(A\#1)}$, $R01A_{(A\#3,S\#11)}$, $A07A_{(S\#11)}$, $C09A_{(A\#7,S\#14)}$
	MoleRec	0.0764	0.2414	21	5	7: $N05C_{(A\#2,S\#5)}$, $A12B_{(A\#1)}$, $N05B_{(A\#2)}$, $R03A_{(A\#1)}$, $R01A_{(A\#3,S\#11)}$, $A07A_{(A\#0,S\#11)}$, $C09A_{(A\#7,S\#14)}$
	4SDrug	0.0439	0.2078	19	7	3: $R03A_{(A\#1)}$, $A07A_{(S\#9)}$, $A12B_{(A\#1)}$

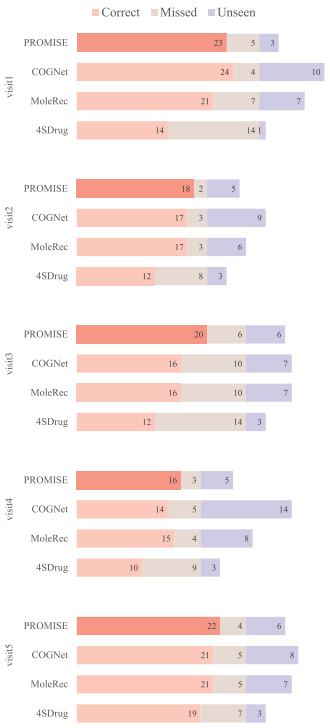


Fig. 13. Example of medication recommendations for patient B in MIMIC-III with multiple visits.

left side details the medication predictions and key indicators in a table format, and the right side offers visual representations of these data points. Within the tables, the notation $*_{(A\#M,S\#N)}$ is used, where $*$ represents a medication incorrectly predicted (listed in the "Unseen" column). This medication has antagonistic interactions (denoted by A) with M correctly predicted medications and synergistic interactions (denoted by S) with N correctly predicted medications.

In addition to prioritizing prediction accuracy, our PROMISE framework also reduces the frequency of unseen medication recommendations compared to other models, thereby enhancing its utility for clinical decision-making. This effectiveness is demonstrated in two key areas: Firstly, compare with other methods, PROMISE identifies a higher number of SDDIs ($*_{(S\#N)}$) among the unseen and correctly predicted medications. This capability suggests that the model is adept at proposing medication combinations that could be beneficial for patient treatment, emphasizing its potential to uncover synergistic medication pairs that could improve therapeutic outcomes. Secondly, PROMISE records a significantly lower number of ADDIs ($*_{(A\#M)}$) between the unseen and correctly predicted medications. This observation underscores the model's effectiveness in avoiding medication combinations that could be detrimental to patient care, thereby enhancing the safety profile of the treatment regimen. These attributes are achieved while maintaining higher SDDI rates and lower ADDI rates, illustrating the balanced approach of PROMISE in optimizing both the effectiveness and safety of its medication recommendations. Moreover, the predictive accuracy of our framework shows marked improvement over successive patient visits. This pattern, corroborated by case studies and previous experimental results, reinforces the conclusion that our framework successfully navigates the dual challenges of efficacy and safety in its medication recommendations.

5.7. Theoretical and practical implications

In this study, we introduce PROMISE, a multimodal representation learning framework designed to enhance medication recommendation tasks by addressing the complexities inherent in Electronic Health Records (EHRs). PROMISE employs an integrative approach, combining various information processing methodologies to develop a comprehensive patient health profile, from local to global perspectives. Initially, PROMISE focuses on modality-specific local representation learning, utilizing multi-source domain knowledge to capture local state representation infused with structural and semantic information. Subsequently, in the multimodal global representation learning stage, it leverages patient trajectory similarities to construct a holistic patient representation. PROMISE learns comprehensive patient representations for drug recommendation by correlating multimodal representations and maintaining single-modality capabilities through cross-modal and modality-specific pre-trained loss functions with information constraints.

Experimental results validated on publicly available datasets MIMIC-III and MIMIC-IV indicate that PROMISE excels in delivering accurate and personalized medication recommendations. Accurate medication recommendation holds substantial practical

importance, primarily by ensures patients receive treatments tailored to their conditions, enhancing treatment efficacy and reducing risks of adverse reactions and complications. Furthermore, it facilitates the formulation of more precise and efficient treatment plans by physicians, contributing to the advancement of personalized medicine. Additionally, PROMISE, beyond being tailored to address a specific challenge within the medical and health sector, encompasses core technologies and methodologies applicable across various tasks. Notably, PROMISE possesses capabilities in multi-modal data integration, extending its utility beyond medication recommendation to encompass diagnosis and risk prediction among other health-related task. Moreover, the local-to-global framework is adept at processing time-series data, rendering it suitable for applications in financial analysis and weather forecasting and other fields. In summary, PROMISE not only excels in its initial field of application but also offers novel opportunities and adaptability for interdisciplinary applications, underscoring the expansive potential of modern technology in addressing a diverse array of challenges.

5.8. Limitation and future research

In this section, we discuss the limitations of our framework and propose future research recommendations.

Limitations: *(1) Temporal Dynamics of Patient Visits:* Following the current research, our framework presently defines each patient visit based solely on the sequence of visits without considering the temporal intervals between these visits. This approach overlooks the potential impact of varying time gaps on patient condition progression and treatment outcomes, which could provide significant insights into patient health trajectories.

(2) Semantic Coherence in Multimodal Frameworks: Within the multimodal framework, textual modality information is generated from the structured EHR data. This process may result in a loss of semantic coherence due to the structured nature of the tabular data, potentially affecting the model's ability to accurately interpret and utilize semantic information.

(3) Narrow Scope of Downstream Tasks: The focus of our framework is limited to a single downstream task — medication recommendation. This singular focus restricts the framework's applicability and overlooks the potential benefits of applying the model to a broader spectrum of clinical prediction tasks, which could enrich the model's utility and relevance in healthcare settings.

Future research should explore several avenues to address current limitations and enhance the framework's efficacy in clinical settings:

(1) Considering the impact of temporal intervals between visits on model performance. The generative models could be developed to produce EHR sequence, thereby mitigating this issue. Such models would simulate realistic patient records, incorporating temporal dynamics to provide a more accurate reflection of patient health trajectories.

(2) Exploring diverse generative approaches could significantly enhance the quality of embeddings used within the framework. Techniques such as employing prompt templates or harnessing the capabilities of large language models could yield embeddings that capture a deeper semantic understanding of the medical text, thus improving the model's ability to interpret and analyze clinical data.

(3) Extending the application of the framework to include a wider variety of clinical prediction tasks, such as diagnostic prediction and relapse prediction, would significantly increase its utility. By broadening the scope of the framework to encompass multiple aspects of patient care and treatment outcomes, the model could provide more comprehensive support for clinical decision-making processes.

Moreover, specific areas offer opportunities for methodological innovation:

(4) In the realm of encoding medication molecular structures, where SMILES notation is commonly used Weininger (1988), innovative methods like DeepSMILES offer an enhanced approach. DeepSMILES (O'Boyle & Dalke, 2018), a variant of SMILES, simplifies the encoding process, making it more suitable for deep learning applications by streamlining the notation. This simplification could potentially improve the efficiency of encoding molecular structures. Additionally, exploring the use of Motifs in molecular encoding presents another avenue for advancing the representation of molecular structures, potentially offering more nuanced insights into molecular characteristics and interactions.

(5) Regarding the computation of patient similarity, our framework introduces the novel concept of comparing similarities between health trajectories. Innovative methods for measuring similarity, such as calculating the similarity between two subgraphs within the EHR hypergraph structure, could be investigated. Furthermore, the selection of an appropriate distance metric should align with the specific characteristics of the data and the objectives of the application. For instance, in cases where the data dimensionality is high or the scales of different dimensions vary significantly, a standardized metric such as the Mahalanobis distance may be preferred. Conversely, when the primary concern is the directionality of the data rather than the absolute distances, cosine similarity could be more suitable. The choice of distance metrics can significantly influence the insights derived from the data. As such, exploring different metrics based on the unique requirements of the research question and the properties of the data could be a fruitful direction for future research. This approach allows for a more tailored analysis that could reveal deeper insights into patient health trajectories and disease progression.

These future directions aim not only to overcome the current framework's limitations but also to harness advanced computational techniques and broader applicability across clinical tasks, enhancing the framework's effectiveness and utility in healthcare analytics.

6. Conclusion

This paper introduces PROMISE, an innovative framework designed for medication recommendation in healthcare. First, PROMISE integrates multi-source and multimodal domain knowledge to obtain medical code embeddings, which is a critical step to improve the accuracy for healthcare predictions. PROMISE proposed a multimodal framework to obtain patient representations with both structural and semantic information across code-level and text-level learning flows. This is achieved through a local to global learning methodology, ensuring a comprehensive patient representation in each modality. Additionally, PROMISE employs various loss functions in its pre-training stage, notably enhancing robustness by constraining and fusing information from multiple modalities. Experimental results validate the superiority and effectiveness of each component within PROMISE, demonstrating its potential for broad applicability. Looking ahead, we plan to integrate an improved similarity measurement module into the framework and conduct more in-depth analyses based on similar patient trajectories in future works.

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CRediT authorship contribution statement

Jialun Wu: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Xinyao Yu:** Writing – review & editing, Investigation, Conceptualization. **Kai He:** Writing – review & editing, Supervision, Methodology, Formal analysis. **Zeyu Gao:** Writing – review & editing, Methodology, Formal analysis. **Tieliang Gong:** Writing – review & editing, Supervision, Funding acquisition.

Data availability

Data will be made available on request.

References

Ali, Z., Huang, Y., Ullah, I., Feng, J., Deng, C., Thierry, N., et al. (2023). Deep learning for medication recommendation: a systematic survey. *Data Intelligence*, *5*(2), 303–354.

Alsentzer, E., Murphy, J., Boag, W., Weng, W.-H., Jindi, D., Naumann, T., et al. (2019). Publicly available clinical BERT embeddings. In *Proceedings of the 2nd clinical natural language processing workshop* (pp. 72–78).

Cai, D., Qian, S., Fang, Q., Hu, J., & Xu, C. (2023). User cold-start recommendation via inductive heterogeneous graph neural network. *ACM Transactions on Information Systems*, *41*(3), 1–27.

Cai, X., Wang, N., Yang, L., & Mei, X. (2022). Global-local neighborhood based network representation for citation recommendation. *Applied Intelligence: The International Journal of Artificial Intelligence, Neural Networks, and Complex Problem-Solving Technologies*, 1–18.

Cheng, D., Yang, F., Xiang, S., & Liu, J. (2022). Financial time series forecasting with multi-modality graph neural network. *Pattern Recognition*, *121*, Article 108218.

Choi, E., Bahadori, M. T., Schuetz, A., Stewart, W. F., & Sun, J. (2016). Doctor AI: Predicting clinical events via recurrent neural networks. In *Machine learning for healthcare conference* (pp. 301–318). PMLR.

Choi, E., Bahadori, M. T., Song, L., Stewart, W. F., & Sun, J. (2017). GRAM: graph-based attention model for healthcare representation learning. In *Proceedings of the 23rd ACM SIGKDD international conference on knowledge discovery and data mining* (pp. 787–795).

Choi, E., Bahadori, M. T., Sun, J., Kulas, J., Schuetz, A., & Stewart, W. F. (2016). RETAIN: An interpretable predictive model for healthcare using reverse time attention mechanism. In *Advances in neural information processing systems* (pp. 3512–3520).

Davis, J., & Goadrich, M. (2006). The relationship between precision-recall and ROC curves. In *Proceedings of the 23rd international conference on machine learning* (pp. 233–240).

Degen, J., Wegscheid-Gerlach, C., Zaliani, A., & Rarey, M. (2008). On the art of compiling and using 'drug-like' chemical fragment spaces. *ChemMedChem*, *3*(10), 1503.

Er, M. J., Zhang, Y., Wang, N., & Pratama, M. (2016). Attention pooling-based convolutional neural network for sentence modelling. *Information Sciences*, *373*, 388–403.

Etemadi, M., Abkenar, S. B., Ahmadzadeh, A., Kashani, M. H., Asghari, P., Akbari, M., et al. (2023). A systematic review of healthcare recommender systems: Open issues, challenges, and techniques. *Expert Systems with Applications*, *213*, Article 118823.

Goldberger, A. L., Amaral, L. A., Glass, L., Hausdorff, J. M., Ivanov, P. C., Mark, R. G., et al. (2000). PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals. *Circulation*, *101*(23), e215–e220.

Gong, F., Wang, M., Wang, H., Wang, S., & Liu, M. (2021). SMR: medical knowledge graph embedding for safe medicine recommendation. *Big Data Research*, *23*, Article 100174.

Han, K., Wang, Y., Chen, H., Chen, X., Guo, J., Liu, Z., et al. (2022). A survey on vision transformer. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *45*(1), 87–110.

Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, *9*(8), 1735–1780.

Indra, S., Wikarsa, L., & Turang, R. (2016). Using logistic regression method to classify tweets into the selected topics. In *2016 international conference on advanced computer science and information systems* icacsis, (pp. 385–390). IEEE.

Islam, M. M., Nooruddin, S., Karray, F., & Muhammad, G. (2023). Multi-level feature fusion for multimodal human activity recognition in Internet of Healthcare Things. *Information Fusion*, *94*, 17–31.

Ji, S., & Ye, J. (2009). Linear dimensionality reduction for multi-label classification. In *Twenty-first international joint conference on artificial intelligence* (pp. 1077–1082).

Johnson, A. E., Bulgarelli, L., Shen, L., Gayles, A., Shammout, A., Horng, S., et al. (2023). MIMIC-IV, a freely accessible electronic health record dataset. *Scientific Data*, *10*(1), 1.

Johnson, A. E., Pollard, T. J., Shen, L., Li-Wei, H. L., Feng, M., Ghassemi, M., et al. (2016). MIMIC-III, a freely accessible critical care database. *Scientific Data*, *3*(1), 1–9.

Jujjavarapu, C., Suri, P., Pejaver, V., Friedly, J., Gold, L. S., Meier, E., et al. (2023). Predicting decompression surgery by applying multimodal deep learning to patients' structured and unstructured health data. *BMC Medical Informatics and Decision Making*, *23*(1), 2.

Kenton, J. D. M.-W. C., & Toutanova, L. K. (2019). Bert: Pre-training of deep bidirectional transformers for language understanding. *vol. 1*, In *Proceedings of naacL-HLT* (p. 2).

Kingma, D. P., & Ba, J. (2014). Adam: A method for stochastic optimization. arXiv preprint arXiv:1412.6980.

Krioukov, D., Papadopoulos, F., Kitsak, M., Vahdat, A., & Boguná, M. (2010). Hyperbolic geometry of complex networks. *Physical Review E*, *82*(3), Article 036106.

Lakkaraju, H., & Rudin, C. (2017). Learning cost-effective and interpretable treatment regimes. In *Artificial intelligence and statistics* (pp. 166–175). PMLR.

Li, Y., Ma, X., Zhou, X., Cheng, P., He, K., & Li, C. (2021). Knowledge enhanced lstm for coreference resolution on biomedical texts. *Bioinformatics*, *37*(17), 2699–2705.

Li, X., Xu, Y., Cui, H., Huang, T., Wang, D., Lian, B., et al. (2017). Prediction of synergistic anti-cancer drug combinations based on drug target network and drug induced gene expression profiles. *Artificial Intelligence in Medicine*, *83*, 35–43.

Liu, Q., Mao, R., Geng, X., & Cambria, E. (2023). Semantic matching in machine reading comprehension: An empirical study. *Information Processing & Management*, *60*(2), Article 103145.

Ma, F., Chitta, R., Zhou, J., You, Q., Sun, T., & Gao, J. (2017). Dipole: Diagnosis prediction in healthcare via attention-based bidirectional recurrent neural networks. In *Proceedings of the 23rd ACM SIGKDD international conference on knowledge discovery and data mining* (pp. 1903–1911).

Ma, L., Gao, J., Wang, Y., Zhang, C., Wang, J., Ruan, W., et al. (2020). AdaCare: Explainable clinical health status representation learning via scale-adaptive feature extraction and recalibration. In *Proceedings of the AAAI conference on artificial intelligence* (pp. 825–832).

Ma, F., Wang, Y., Xiao, H., Yuan, Y., Chitta, R., Zhou, J., et al. (2019). Incorporating medical code descriptions for diagnosis prediction in healthcare. *BMC Medical Informatics and Decision Making*, *19*(6), 1–13.

Ma, F., You, Q., Xiao, H., Chitta, R., Zhou, J., & Gao, J. (2018). Kame: Knowledge-based attention model for diagnosis prediction in healthcare. In *Proceedings of the 27th ACM international conference on information and knowledge management* (pp. 743–752).

Mei, X., Cai, X., Xu, S., Li, W., Pan, S., & Yang, L. (2022). Mutually reinforced network embedding: An integrated approach to research paper recommendation. *Expert Systems with Applications*, *204*, Article 117616.

Mu, Y., & Wu, Y. (2023). Multimodal movie recommendation system using deep learning. *Mathematics*, *11*(4), 895.

Muhammad, G., Alshehri, F., Karray, F., El Saddik, A., Alsulaiman, M., & Falk, T. H. (2021). A comprehensive survey on multimodal medical signals fusion for smart healthcare systems. *Information Fusion*, *76*, 355–375.

Müller, M. (2007). Dynamic time warping. In *Information retrieval for music and motion* (pp. 69–84). Springer.

Niwattanakul, S., Singthongchai, J., Naenudorn, E., & Wanapu, S. (2013). Using of Jaccard coefficient for keywords similarity. In *Proceedings of the international multiconference of engineers and computer scientists* (pp. 380–384).

O'Boyle, N., & Dalke, A. (2018). DeepSMILES: an adaptation of SMILES for use in machine-learning of chemical structures.

Read, J., Pfahringer, B., Holmes, G., & Frank, E. (2011). Classifier chains for multi-label classification. *Machine Learning*, *85*, 333–359.

Shang, J., Ma, T., Xiao, C., & Sun, J. (2019). Pre-training of graph augmented transformers for medication recommendation. In *28th international joint conference on artificial intelligence* IJCAI 2019, (pp. 5953–5959). International Joint Conferences on Artificial Intelligence.

Shang, J., Xiao, C., Ma, T., Li, H., & Sun, J. (2019). GAMENet: Graph augmented memory networks for recommending medication combination. In *Proceedings of the AAAI conference on artificial intelligence* (pp. 1126–1133).

Solt, I., & Tikk, D. (2009). Yet another rule-based approach for extracting medication information from discharge summaries. In *Proceedings of the third i2b2 workshop on challenges in natural language processing for clinical data*.

Sun, H., Xie, S., Li, S., Chen, Y., Wen, J.-R., & Yan, R. (2022). Debiased, longitudinal and coordinated drug recommendation through multi-visit clinic records. *Advances in Neural Information Processing Systems*, *35*, 27837–27849.

Suo, Q., Ma, F., Yuan, Y., Huai, M., Zhong, W., Zhang, A., et al. (2017). Personalized disease prediction using a cnn-based similarity learning method. In *2017 IEEE international conference on bioinformatics and biomedicine* BIBM, (pp. 811–816). IEEE.

Tan, Y., Kong, C., Yu, L., Li, P., Chen, C., Zheng, X., et al. (2022). 4Sdrug: Symptom-based set-to-set small and safe drug recommendation. In *Proceedings of the 28th ACM SIGKDD conference on knowledge discovery and data mining* (pp. 3970–3980).

Tang, R., Yang, C., & Wang, Y. (2023). A cross-domain multimodal supervised latent topic model for item tagging and cold-start recommendation. *IEEE MultiMedia*.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., et al. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, *30*.

Velickovic, P., Cucurull, G., Casanova, A., Romero, A., Lio, P., Bengio, Y., et al. (2017). Graph attention networks. *Statistics*, *1050*(20), 10–48550.

Wang, N., Cai, X., Yang, L., & Mei, X. (2022). Safe medicine recommendation via star interactive enhanced-based transformer model. *Computers in Biology and Medicine*, *141*, Article 105159.

Wang, Y., Chen, W., Pi, D., & Yue, L. (2021). Adversarially regularized medication recommendation model with multi-hop memory network. *Knowledge and Information Systems*, *63*(1), 125–142.

Wang, L., Zhang, W., He, X., & Zha, H. (2018). Personalized prescription for comorbidity. In *Database systems for advanced applications: 23rd international conference, DASFAA 2018, gold coast, QLD, Australia, May 21-24, 2018, proceedings, part II 23* (pp. 3–19). Springer.

Wang, S., Zhang, X., Wang, Y., & Ricci, F. (2022). Trustworthy recommender systems. *ACM Transactions on Intelligent Systems and Technology*.

Weininger, D. (1988). SMILES, a chemical language and information system. 1. introduction to methodology and encoding rules. *Journal of Chemical Information and Computer Sciences*, *28*(1), 31–36.

Wishart, D. S., Feunang, Y. D., Guo, A. C., Lo, E. J., Marcu, A., Grant, J. R., et al. (2018). DrugBank 5.0: a major update to the DrugBank database for 2018. *Nucleic Acids Research*, *46*(D1), D1074–D1082.

Wu, J., Dong, Y., Gao, Z., Gong, T., & Li, C. (2023). Dual attention and patient similarity network for drug recommendation. *Bioinformatics*.

Wu, J., He, K., Mao, R., Li, C., & Cambria, E. (2023). MEGACare: Knowledge-guided multi-view hypergraph predictive framework for healthcare. *Information Fusion*, *100*, Article 101939.

Wu, J., Qian, B., Li, Y., Gao, Z., Ju, M., Yang, Y., et al. (2022). Leveraging multiple types of domain knowledge for safe and effective drug recommendation. In *Proceedings of the 31st ACM international conference on information & knowledge management* (pp. 2169–2178).

Wu, R., Qiu, Z., Jiang, J., Qi, G., & Wu, X. (2022). Conditional generation net for medication recommendation. In *Proceedings of the ACM web conference 2022* (pp. 935–945).

Wu, J., Zhang, R., Gong, T., Zhang, H., Wang, C., & Li, C. (2021). A personalized diagnostic generation framework based on multi-source heterogeneous data. In *2021 IEEE international conference on bioinformatics and biomedicine* BIBM, (pp. 2096–2103). IEEE.

Yang, C., Xiao, C., Glass, L., & Sun, J. (2021). Change matters: Medication change prediction with recurrent residual networks. In *30th international joint conference on artificial intelligence* IJCAI 2021, (pp. 3728–3734). International Joint Conferences on Artificial Intelligence.

Yang, C., Xiao, C., Ma, F., Glass, L., & Sun, J. (2021). SafeDrug: Dual molecular graph encoders for recommending effective and safe drug combinations. In *30th international joint conference on artificial intelligence* IJCAI 2021, (pp. 3735–3741). International Joint Conferences on Artificial Intelligence.

Yang, N., Zeng, K., Wu, Q., & Yan, J. (2023). Molerec: Combinatorial drug recommendation with substructure-aware molecular representation learning. In *Proceedings of the ACM web conference 2023* (pp. 4075–4085).

Yi, K., Zhang, Q., Cao, L., Wang, S., Long, G., Hu, L., et al. (2023). A survey on deep learning based time series analysis with frequency transformation. CoRR, abs/2302.02173.

Yin, C., Liu, R., Zhang, D., & Zhang, P. (2020). Identifying sepsis subphenotypes via time-aware multi-modal auto-encoder. In *Proceedings of the 26th ACM SIGKDD international conference on knowledge discovery & data mining* (pp. 862–872).

Zhang, Y., Chen, R., Tang, J., Stewart, W. F., & Sun, J. (2017). LEAP: Learning to prescribe effective and safe treatment combinations for multimorbidity. In *Proceedings of the 23rd ACM SIGKDD international conference on knowledge discovery and data mining* (pp. 1315–1324).

Zhang, P., Wang, F., Hu, J., & Sorrentino, R. (2014). Towards personalized medicine: leveraging patient similarity and drug similarity analytics. *AMIA Summits on Translational Science Proceedings*, *2014*, 132.

Zhao, W., Jiang, X., Wang, K., Sun, X., Hu, G., & Xie, G. (2020). Construction of guideline-based decision tree for medication recommendation. *Studies in Health Technology and Informatics*.

Zhu, Y., Lin, Q., Lu, H., Shi, K., Liu, D., Chambua, J., et al. (2021). Recommending learning objects through attentive heterogeneous graph convolution and operation-aware neural network. *IEEE Transactions on Knowledge and Data Engineering*.